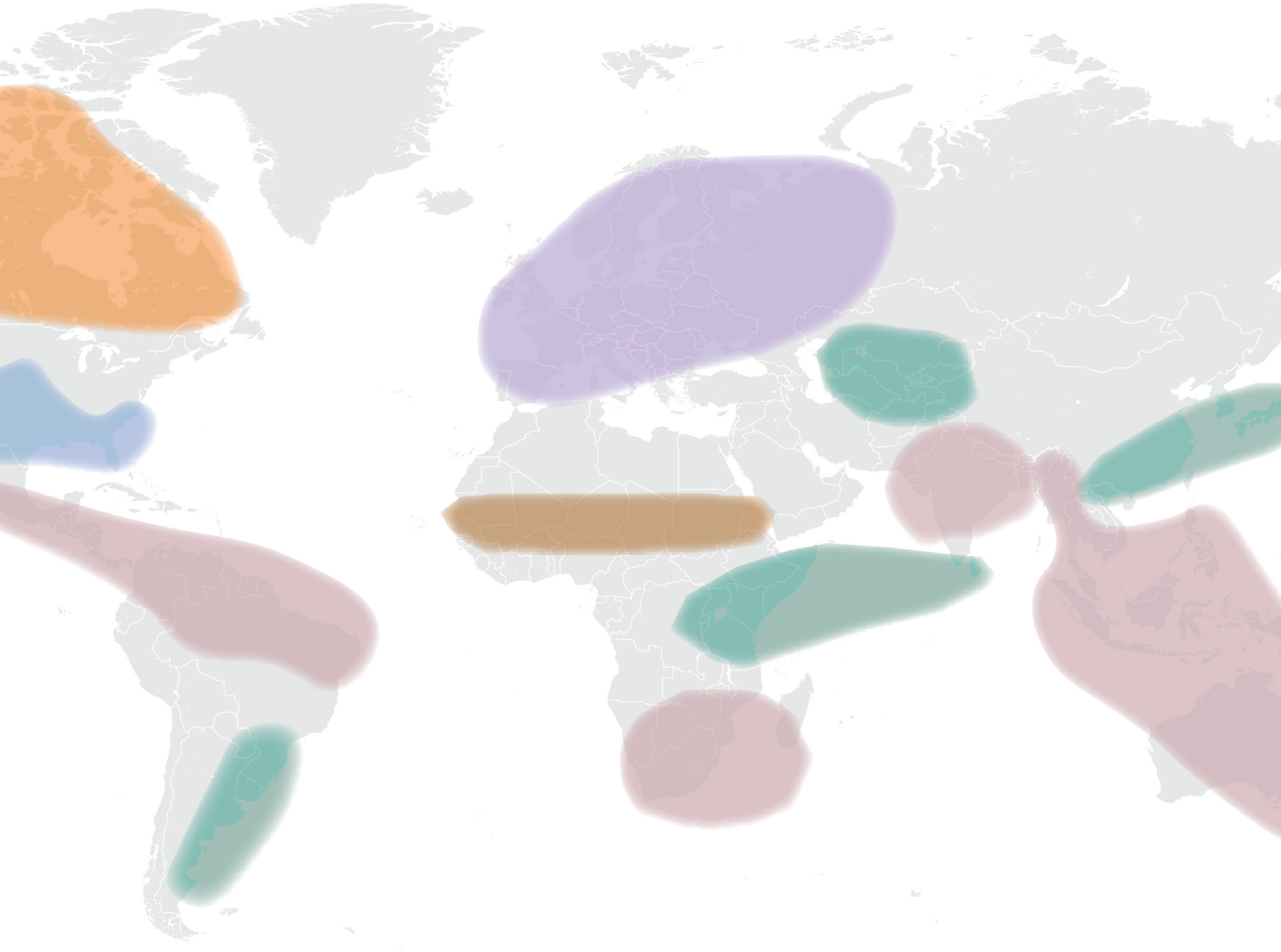
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Joint Research Centre

From forecast to risk: an anticipatory analysis of the 2026-27 El Niño

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Abstract

The occurrence of the 2026-27 El Niño event is virtually certain based on all the seasonal forecast modelling systems contributing to the Copernicus Climate Change Service. This event is also expected to reach very high intensity, with forecasts indicating a high likelihood of a very strong event that may even evolve into an unprecedented one. Drawing on multi-century climate model simulations, seasonal forecasting systems, and socio-economic data, this report analyses potential impacts across several dimensions and the regional responses to different El Niño categories.

It is important to remark that the effects of El Niño build on top of the ongoing global warming, existing vulnerabilities and crises. On the climate dimension, the analysis identifies hot-spots of temperature and water-related anomalies and extremes in America, Africa, Asia, and Oceania varying according to the intensity of El Niño. It also points to nonlinear response to a possible unprecedented event in regions such as Europe, where warmer conditions may occur in autumn and intensify in spring 2027. El Niño can cause severe, systemic impacts across sectors and natural systems due to the concurrent and compounding risks it triggers by, globally, amplifying extremes and anomalous climate conditions. Since agriculture may represent one of the main transmission channels of socio-economic impacts, the response of global food prices is here analysed. The estimated outlook for the main commodity crops is characterised by high uncertainties, although overall price increases are estimated for durum wheat and (with less clear tendency) for maize.

On population exposure to climate hazards, the impacts scale sharply with El Niño intensity, with Oceania, Sub-Saharan Africa, Eastern and Southeastern Asia, and Latin America facing the greatest burdens, particularly among vulnerable demographic groups. On human mobility, it is difficult to make accurate predictions as past evidence points to systematic undercounting of drought-driven displacement and the critical but invisible risk of involuntary immobility among the poorest populations. Across all dimensions, the predictability of El Niño and the coherent signal provided by the current seasonal forecasts represent key assets for anticipatory actions and provide a valuable window to prepare and reduce risks.

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Executive summary

Policy context

On 2 June 2026, the World Meteorological Organisation issued a call to prepare for El Niño given the estimated likelihood of occurrence (80% for June-August and 90% to continue at least until November, based on the WMO global producing centres for seasonal predictions and expert assessment). ENSO (El Niño Southern Oscillation) represents one of the most predictable sources of climate variability, arising from a coupled interaction between the tropical Pacific Ocean and the over-lying atmosphere. The effects of ENSO are felt globally and lead to increased hazards in well-known regions of the world. As a reliable "early warning", the multi-month lead time provided by ENSO forecasts creates a genuine opportunity for governments, international organisations, and humanitarian actors for anticipatory action. If governance and financing is mobilised in time, it allows to pre-position resources, adjust food and water security strategies, and protect the most vulnerable populations. Yet this opportunity is frequently missed: response efforts tend to be reactive, triggered when displacement or harvest failures become tangible rather than by forecast confirmation. Concerning this 2026/27 event, some anticipatory measures have been already put in place, such as UNDRR (United Nations Office for Disaster Risk Reduction) El Niño guidelines, IFRC (International Federation of Red Cross and Red Crescent Societies) early-action financing, and regional water-rationing plans in several countries. As El Niño impacts intensify in a warming climate, the gap between what science can predict and what policy delivers grows ever more consequential, particularly for the regions least equipped to absorb the shocks.

As early-warning is essential to effectively reduce risks associated with ENSO, it is worth to highlight the role of Copernicus Services, such as the Emergency Management Service and the Climate Change Service, to anticipate impacts and provide actionable data and information. Strengthening these tools for flood, drought, heatwave and wildfire monitoring, and sectoral impact analysis based on them will be crucial in the coming months. The same holds for tools such as the Global Disaster Awareness and Coordination System and the Anomaly Hotspots of Agricultural Production. In combination with our INFORM assessment of current humanitarian vulnerability and risk, these data and information can support the EU Emergency Response Coordination Centre to anticipate response and actions. More importantly, accelerating Global Gateway initiatives, such as SEWA (Strengthening Early Warning in Africa) and StormWatch, can support – by building on the EU's Copernicus programme – the development of early warning capacity in the most affected regions like Africa and the Caribbean.

Key conclusions

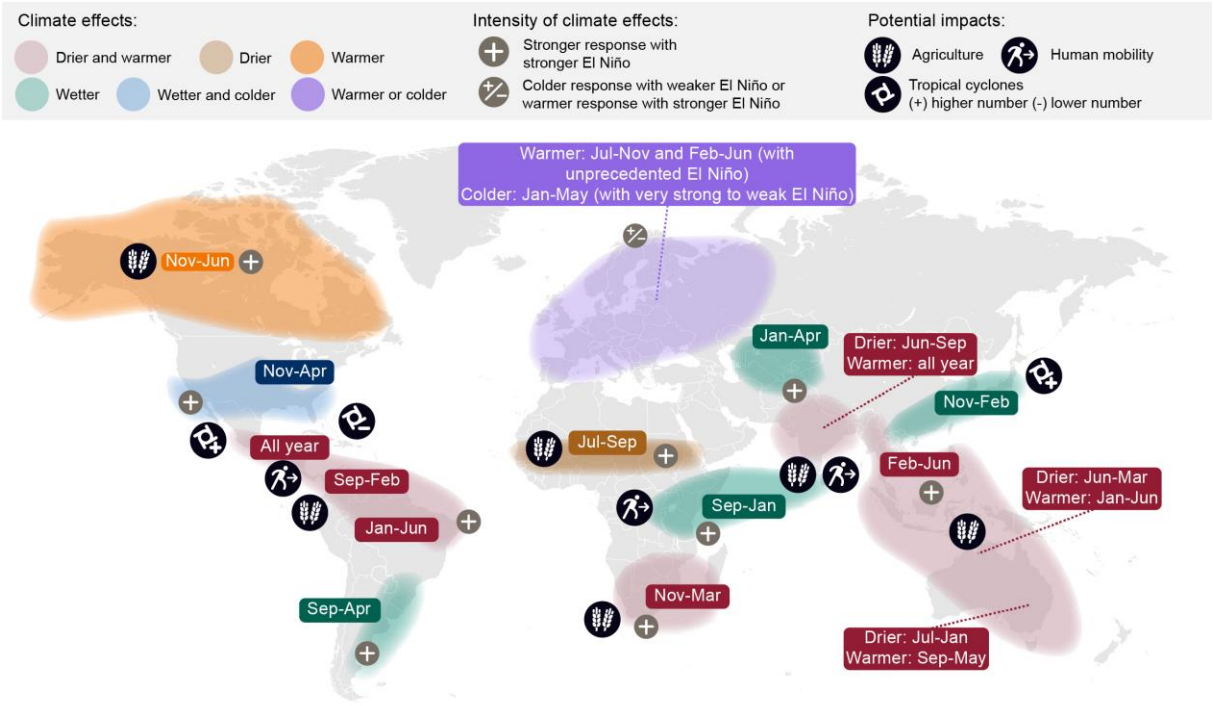
Seasonal forecasts contributing to the Copernicus Climate Change Service point to a virtually certain 2026/27 El Niño event that may be very strong and even evolve into an unprecedented one. El Niño poses compounding risks across multiple dimensions that are uneven in their geography but interconnected in their consequences. Importantly, El Niño-related shocks may coincide with pre-existing vulnerabilities including fragile supply chains, elevated energy and fertilizer prices, humanitarian stress and geopolitical tensions. Here, we focus on population exposure to climate hazards, food prices, human mobility, and humanitarian risks, while also considering their broader macroeconomic and financial risk transmission channels. Heat, drought, floods and over-wet conditions emerge as the dominant hazards, disproportionately affecting large regions of America, Asia, Africa, and Oceania, with agricultural systems, public health, and vulnerable demographics bearing the greatest burden. Population exposure to these hazards scales sharply with the El Niño

intensity, implying that the difference between a strong and an unprecedented El Niño is not merely meteorological, but it translates directly into hundreds of millions of additional people at risk. The overall food price outlook is somewhat heterogeneous and characterised by higher uncertainties. However, durum wheat stands out as a clear pressure point and maize shows a (less clear) price increase tendency. Regional food security assessments, anyhow, remain essential given the limits and the uncertainties of global price assessments. On mobility, the evidence cautions against oversimplified narratives: drought-driven movement is systematically undercounted, immobility in the poorest areas is a silent crisis, and most displacement remains local. Across all these dimensions, the central policy conclusion is the same: the forecast lead time is an asset that must be acted upon now, before impacts cascade into emergencies that are far costlier to manage than to anticipate.

Main findings

Map with key highlights

Figure 1. Major impacts and main climate effects of El Niño events.



Sources: JRC analysis and Lenssen et al. (2020).

Impacts of El Niño on the climate system, plausible scenarios and forecasts

— Temperature anomalies and extremes

- Extreme heat across the tropics and subtropics (sub-Saharan Africa, South/Southeast Asia, Australia, America) will build from September, peak in December–February and persist through May.
- The more intense the El Niño, the earlier and longer the heat anomalies will last. Under an unprecedented event, significant heat may emerge as early as June.

    - Northern Eurasia and parts of North America may experience colder-than-normal conditions, mostly around the boreal winter months.
    - In case of an unprecedented El Niño warmer conditions are expected to occur in late 2026 and intensify in spring 2027 over the entire Eurasia region.

— Anomalous water conditions and extremes

    - Drought patterns are strongly regional and tend to outlast the event itself: agricultural drought and food insecurity and water stress can persist months after the El Niño peak.
    - Australia and Southeast Asia may face multi-season drying, worsening with event intensity.
    - Southern Africa, Central America, northern South America, the Sahel, and India may face significant drought events during their respective rainy season.
    - East Africa, the Middle East, and Central Asia may face well-above-normal rainfall.
    - Southeastern South America (Argentina, Uruguay, southern Brazil) and the western United States may face enhanced rainfall from boreal autumn through spring.

— El Niño will shift tropical storm activity: The expected number of Pacific hurricane formation rises along Central America/Mexico, while it is suppressed in the Atlantic/Caribbean, and western Pacific expected number of typhoon likelihood increases.

Crops and food prices

— The assessment of El Niño effects on global prices of the main commodity crops reveals high uncertainties for some crops and limitations (especially related to the possibility of an unprecedented event). Regional assessments and monitoring remain, therefore, essential.

— Global rice prices are expected to slightly decline in the first phase of El Niño and increase later on in all El Niño categories.

— Global soybean and hard red winter wheat prices are expected to decline in all El Niño categories.

— Durum wheat is the clearest concern as global price is expected to increase in the coming months in all El Niño categories.

— Global maize price is expected to slightly increase under all El Niño categories except the most extreme. The identified response under an unprecedented El Niño must be interpreted with caution due to models' limitations.

Population exposure to hazards and humanitarian risks

— El Niño events significantly increase population exposure to four key climate hazards: high precipitation, drought, high temperature and low temperature extremes — with severity scaling from weak to unprecedented El Niño events except for low temperature extremes.

— Globally, over 20% of the population may be exposed to low temperature extremes, while drought and high temperature exposure will also rise sharply with stronger El Niño.

— Oceania may face the highest proportional hazard exposure, particularly during the most extreme El Niño categories.

— Pacific Island indigenous communities may be disproportionately affected, relying on traditional livelihoods. The 2015–16 El Niño caused severe water shortages and food insecurity.

— El Niño may amplify existing vulnerabilities in Sub-Saharan Africa related to rapid population growth, high poverty, and young demographics and undernutrition during droughts.

— Eastern and Southeastern Asia, with 2.3 billion people, may see up to 500 million exposed to heatwaves during strong events, with rural communities heavily disrupted by drought-related crop failures.

— In Latin America, over 60% urban populations in informal settlements may face amplified flood and heatwave risks; droughts may also push rural-to-urban migration (1.4M were displaced e.g. in 2018–19).

— Stronger El Niño events consistently drive more prominent heat and drought exposure globally, with major implications for food security and public health.

— Humanitarian risk for the next six months due to dry conditions may be concentrated in northern South America, the Caribbean, southern Central America, Australia, Kazakhstan, and Southeast Asia — with Bolivia carrying the highest compound warning signal, driven by severe drought and high inflation.

— Of the top ten countries that may face the strongest dry conditions, only Ecuador, Venezuela, and Haiti also register high humanitarian warning levels, reflecting the role of structural vulnerabilities in determining overall crisis risk.

— Nigeria and Central African Republic are among the countries with the highest global warning levels, where El Niño related impacts may overlap with conflict, displacement, food insecurity and economic stress, further compounding already severe crisis conditions in these countries.

— High river flow anomalies in the coming months are predicted mainly in Southern Africa and Western Asia, with countries like Iraq or Tanzania show hydrological stress that intersects with existing socio-economic fragility.

Human mobility

— Predicting the effects of the 2026/27 El Niño event on human mobility is not trivial as past evidence has already shown. Floods/storms cause rapid and visible displacement; drought instead drives gradual movement, and it is systematically undercounted in official statistics.

— Climate-related mobility is overwhelmingly internal and short-distance. Evidence does not support long-distance mass migration induced by events such as El Niño.

— Immobility is a hidden risk: in the poorest areas, drought depletes assets until people cannot move — a humanitarian crisis invisible to displacement data.

— Recording bias distorts the picture. The drought-dominated 2015–16 El Niño showed below-average displacement figures; 2024's flood-dominated event set a record (~45.8 million). The data reflects what is measurable, not what is impactful.

— 2026 high-risk regions such as Southern Africa, the Horn of Africa, Central Africa (floods and conflict), and the Central American Dry Corridor[1] may face compounded vulnerabilities, often layered on still-unresolved 2023–24 impacts and crises.

Systemic Risks

— El Niño can lead to significant and enduring effects on various sectors, with impacts differing based on phase, region, and sectoral exposure.

— El Niño affects the economy primarily via agricultural production, food commodity prices, inflation, and long-term GDP growth. Major El Niño events have historically reduced economic growth, with substantial global income losses.

— Economic impacts are not confined to the event period, as losses can persist for years after the initial shock produced by El Niño.

— El Niño can disrupt infrastructure, energy systems, and global trade, as seen in drought-related low water levels affecting shipping routes like the Panama Canal.

— Drought conditions amplified or caused by El Niño can reduce hydropower generation, increasing energy stress and production costs, affecting import-dependent economies and small island states.

— El Niño can destabilise banking sectors through credit-risk channels, especially in climate-sensitive regions such as Latin America, the Caribbean, and Sub-Saharan Africa.

— The effects on banking and economic stability are exacerbated in countries already facing high debt, limited public resources, or financial-market instability.

Multi-lateral NGO responses, reported news, planned measures

— Global coordination: the UN, UNDRR, and IFRC are leading multilateral responses, with the IFRC activating Early Action Protocols across Latin America and the Caribbean ahead of predicted 2–3 °C Pacific temperature spikes.

— Americas: Panama has enacted emergency water rationing heavily impacting Canal transit; Colombia has activated drought, wildfire and water scarcity preparedness measures; Belize has launched anticipatory actions against below-average rainfall linked to ENSO patterns.

— Africa and Asia: Zimbabwe may face a high probability of El Niño-driven drought and poor harvests in 2026/27; ASEAN countries — particularly Mekong nations, Nepal, and Malaysia — may face severe hydropower and agricultural disruption, compounded by fertilizer shortages from the Persian Gulf conflict.

[1] The Dry Corridor or Central American Dry Corridor (CADC) extends from southern Mexico to Panama along the Pacific Coast.

— South and Southeast Asia: India's monsoon is forecast below-normal, with drought risk in the north and west and flooding in the southeast; Vietnam warns of intense heatwaves, 25–50% rainfall reduction, and flash flood risk; the Philippines anticipates flooding during the western monsoon season followed by drought into early 2027.

— Oceania and Europe: New Zealand may face drought, extreme heat, and wildfire risk affecting groundwater and agriculture; the UK's Committee on Climate Change has legally advised rapid infrastructure upgrades against heatwaves, floods, and wildfires amplified by a high-intensity El Niño.

— Policy response: the UN Secretary-General Guterres issued a global directive to accelerate the transition away from fossil fuels, warning that El Niño acts as a "threat multiplier" for climate extremes.

Related and future work

Critically, forecasts issued before summer in the northern hemisphere still carry meaningful uncertainty around the event's ultimate magnitude, and the picture will sharpen considerably with updates expected in July, making continuous monitoring of ENSO forecasts a policy imperative, as the scale of preparedness and resource mobilisation required depends directly on how the phenomenon evolves.

Quick guide

This document provides a comprehensive anticipatory analysis of the predicted 2026/27 El Niño event across several dimensions with the main aim of providing scientific evidence for policy recommendations emphasising the importance of anticipatory action. It opens with a general brief introduction to El Niño phenomenon in Section 1. Section 2 provides an analysis of what to expect in terms of temperature, drought, precipitation, floods and tropical storm patterns by region and event intensity. Section 3 analyses the latest forecasts. Section 4 then examines food price impacts, with scenario-based projections for maize, rice, soybeans, and wheat. Section 5 quantifies population exposure to climate hazards across key world regions and presents the countries with highest risk of humanitarian impacts due to El Niño over the next six months. Section 6 presents an analysis of potential human mobility related to El Niño. Finally, section 7 provides an overview of collected NGO (non-governmental organisation), national and international responses and action plans underway to mitigate the effects of the phenomenon, together with a collection of news on the forecasted El Niño event and its possible impacts.

1. Introduction

The El Niño Southern Oscillation (ENSO) is among the most prominent natural, large-scale climate pattern on the planet (Cai et al., 2021; McPhaden et al., 2006), arising from a coupled interaction between the tropical Pacific Ocean and the over-lying atmosphere (Bjerknes, 1969; Bjerknes, 1966). During an ENSO event the sea-surface temperature (SST) in the central and eastern equatorial Pacific deviates by several degrees from its long-term mean (Trenberth, 1997), while the atmospheric pressure gradient between the western Pacific and the Indian-Ocean sector of the atmosphere (the Southern Oscillation) reverses sign (Bjerknes, 1969; Bjerknes, 1966). These simultaneous oceanic and atmospheric anomalies constitute a single, self-reinforcing system rather than independent phenomena (Santoso et al., 2017; Bjerknes, 1969).

ENSO cycles through three broadly defined phases. In the warm phase, El Niño, weakened easterly trade winds allow warm water that normally piles up in the western Pacific to flow eastward and suppressing upwelling of cold, nutrient-rich water along the South American coast (Wyrtki, 1975; Bjerknes, 1969). The atmospheric response includes reduced cloudiness and rainfall over Indonesia and enhanced convection over the central-eastern Pacific (Santoso et al., 2017; Bjerknes, 1969). The cool phase, La Niña, is essentially the opposite: stronger trade winds intensify upwelling, and produce cooler SSTs, leading to increased rainfall over the western Pacific (Santoso et al., 2017). Between these extremes lies the neutral state, when SSTs and atmospheric pressure are close to their climatological averages (Trenberth, 1997).

Because ENSO modulates the global atmospheric circulation, its influence extends far beyond the Pacific basin (Yeh et al., 2018; McPhaden et al., 2006, Bjerknes, 1966). Typical teleconnections include altered jetstream patterns that raise winter precipitation in the southern United States and northern Mexico and suppress it in the northcentral region during El Niño, while La Niña produces the converse pattern (Ropelewski and Halpert 1987; Ropelewski and Halpert, 1986). Similar anomalies reduce monsoon intensity in South Asia and Central America (Webster et al., 1998), affect tropical cyclone activity in the Atlantic (Gray et al., 1984). Although largely uncertain, El Niño events may have had impacts on European countries (Brönnimann, 2007), particularly in terms of extreme weather events and climate-related disasters. For instance, the 1997-1998 El Niño event possibly contributed to severe flooding in several European countries, including the UK, France, and Italy (IPCC, 2019). Similarly, the 2015-2016 El Niño event resulted in droughts and heatwaves in several European countries, including Spain, Portugal, and Greece.

The relatively high predictability of ENSO — forecasts can be issued several months in advance (McPhaden et al., 2006; Latif et al., 1998) — makes it a cornerstone of seasonal climate prediction and a critical factor in planning for agriculture, water resources and disaster risk management worldwide (IIzumi et al., 2014; McPhaden et al., 2006).

ENSO exemplifies a large-scale, coupled ocean–atmosphere system whose phases are defined by measurable SST and pressure anomalies, underpinned by well-established physical feedbacks, and whose impacts are felt globally, underscoring its scientific and societal relevance (Santoso et al., 2017; McPhaden et al., 2006).

The US National Oceanic and Atmospheric Administration (NOAA) has confirmed the emergence of an El Niño event, characterised by warmer-than-average sea surface temperatures in the central and eastern Pacific Ocean. According to the NOAA Climate Prediction Center, the current El Niño event is expected to persist through the winter months, potentially leading to significant climate-related

impacts[2]. The World Meteorological Organization (WMO) reported in late April a high likelihood of an El Niño event developing in mid-2026 and on 2 June 2026[3] pointed to 80% probability of occurrence in June-August with 90% probability of continuing at least until November 2026.

ENSO can generate large and persistent systemic impacts, although effects differ by phase, region, and sectoral exposure (Dufrénot et al., 2024). It affects the broader economy mainly through agricultural production, food commodity prices, inflation and impact on longer-term GDP (Gross Domestic Product) growth. Major El Niño events have reduced country-level economic growth over extended periods, with estimated global income losses of approximately EUR 3.8 trillion and EUR 5.2 trillion for the 1982–83 and 1997–98 events, respectively (Callahan and Mankin, 2023). Importantly, these impacts are not limited to the event period, as economic losses can continue to accumulate for several years after the initial shock. El Niño can disrupt infrastructure, energy systems and global trade. Drought-related low water levels, such as those observed in the Panama Canal during the 2023–24 event, can restrict shipping and increase transport costs across global supply chains (WMO, 2024). In hydropower-dependent regions, drought conditions can also reduce electricity generation, increasing energy stress and production costs. These disruptions are particularly relevant for import-dependent economies and small island states, where higher energy and food costs can translate into inflationary pressures, fiscal strain and reduced economic resilience.

ENSO can affect banking-sector stability through credit-risk channels (Damette et al., 2026). Droughts, floods and agricultural losses can weaken household and firm incomes, increase non-performing loans and deteriorate bank asset quality, particularly in economies highly exposed to climate-sensitive sectors. These effects appear especially pronounced for El Niño events in Latin America and the Caribbean and Sub-Saharan Africa. The transmission from climate shocks to banking-sector stress is closely linked to worsening borrower repayment capacity following agricultural and broader economic losses. More broadly, these risks can become much more severe when climate shocks hit countries that already face high debt, limited public resources or financial-market instability (Callahan and Mankin, 2023).

Given the latest forecasts, El Niño-related shocks may compound with elevated energy and fertilizer prices, fragile supply chains and geopolitical tensions, increasing the likelihood of compound and non-linear systemic impacts. A plausible transmission pathway would run from droughts, floods and heat stress affecting agricultural production, labour productivity, water availability, hydropower generation and transport systems, to higher food and energy prices, inflationary pressure, fiscal stress and weaker borrower repayment capacity. Particularly exposed regions would likely include parts of Latin America, southern and eastern Africa, South and Southeast Asia, and small islands or import-dependent economies. As for Europe, an unprecedented El Niño could bring more direct physical impacts, especially after summer, through increased heatwave, drought and flood risks affecting agriculture, water availability, energy demand, transport infrastructure and ecosystems. These direct risks would come on top of indirect exposure through commodity-market volatility, supply-chain disruption and humanitarian or development-finance needs.

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2 https://www.cpc.ncep.noaa.gov/products/analysis_monitoring/enso_advisory/ensodisc.shtml (14 May 2026, last accessed: 29 May 2026).

3 https://wmo.int/news/media-centre/wmo-prepare-el-nino?access-token=Y0qE1XztCZT7VZrFwUErPN7veERMVRy3tSP-sw-f_KM.

2. Impacts of El Niño on the climate system and plausible scenarios

Due to the considerable likelihood of experiencing a very strong El Niño event this year (see Section 3), we here adopt the usual El Niño events definition (Table 1) with the addition of two new categories for the most extreme events: super and unprecedented. We here use the NINO3.4 index, which measures sea surface temperature (SST) anomalies in the central and eastern tropical Pacific Ocean, specifically in the region defined by coordinates 5 °N to 5 °S latitude and 120 °W to 170 °W longitude. The index is the most common Oceanic El Niño Index, peaking in the period of maximum strength of El Niño, namely November-January. Since 1950, there has only been one Super El Niño event (2015/2016) and no event has gone beyond the 3.0 °C limit, hence the name unprecedented for the new category.

To get robust understanding of the global effects related to the most extreme categories of El Niño (super and unprecedented), we analysed a simulation covering the last millennium of a global climate model (CESM2, Danabasoglu et al., 2020). This makes possible to sample a considerably long period of stable climate (under pre-industrial conditions) having enough recurrence of El Niño events so to correctly diagnosing the associated climate impacts from the onset in late boreal spring/summer until the decay in spring of the following year.

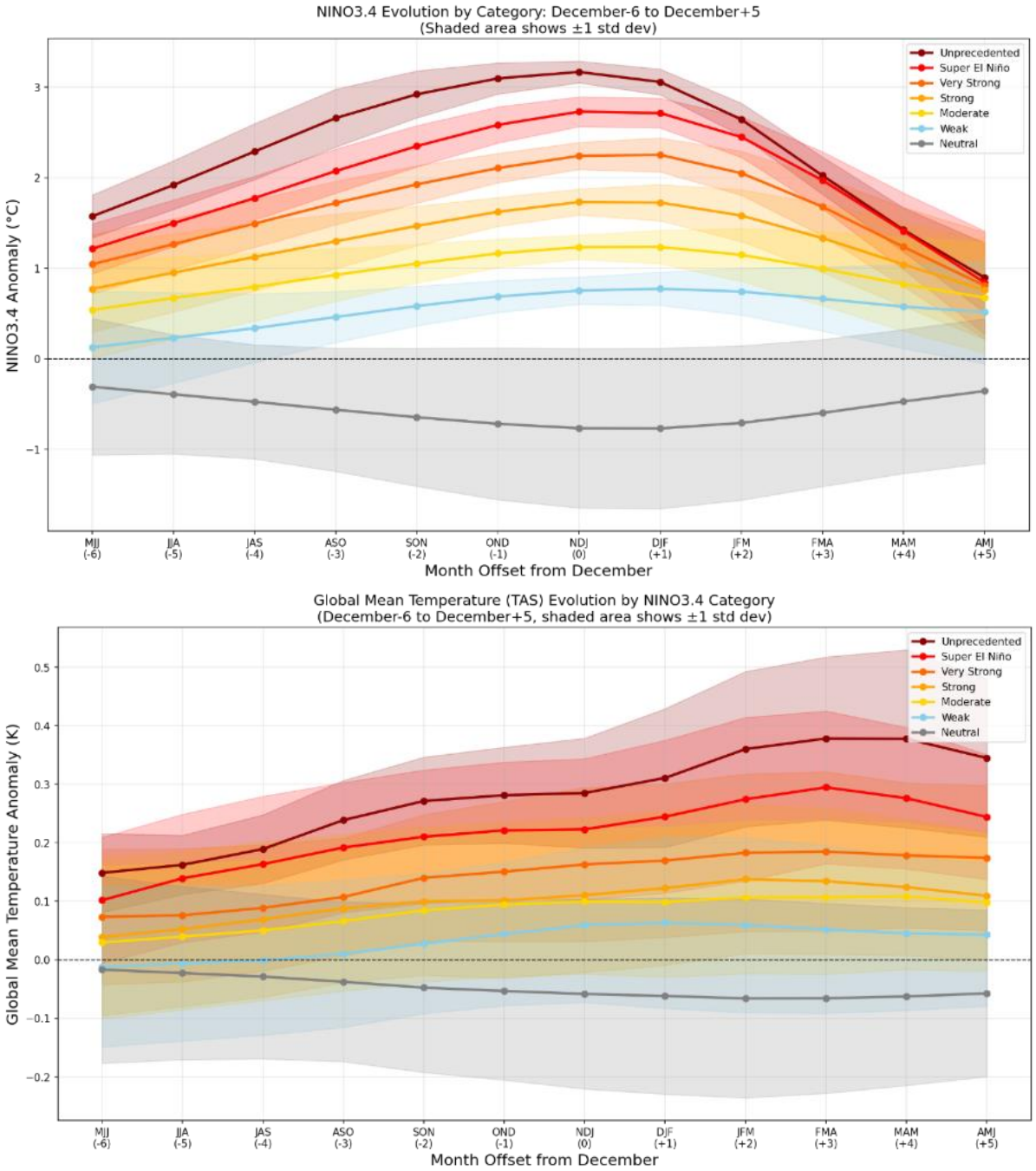
Table 1. Definition of El Niño categories and statistics of El Niño events in the multi-century climate model simulation defined as the 3-month mean from November to January.

El Niño Category	Category Limits (NINO3.4)	Count	Return Period
Unprecedented	> 3.0 °C	10	~1/120 yrs.
Super	2.5-3.0 °C	37	~1/32 yrs.
Very Strong	2.0-2.5 °C	48	~1/25 yrs.
Strong	1.5-2.0 °C	68	~1/18 yrs.
Moderate	1.0-1.5 °C	111	~1/11yrs.
Weak	1.0-0.5 °C	141	~1/9 yrs.
Neutral & La Niña	< 0.5 °C	784	~2/3 yrs.

Source: JRC, based on the CESM2 (Community Earth System Model version 2) pre-industrial model simulation of the last millennium.

Figure 2 (top) shows how the NINO3.4 Index evolves over time for different El Niño intensity categories, from 6 months before to 5 months after the typical time of maximum intensity (i.e. December). This classification is essential for understanding the varying strengths of El Niño events and its typical cycle from its onset around boreal spring/summer until dissipation the following spring. It is worth noting that despite many similarities shared between El Niño events, no events are the same and the impacts can vary from one to the other. All warm categories show a clear rise and fall centred around the November-January period.

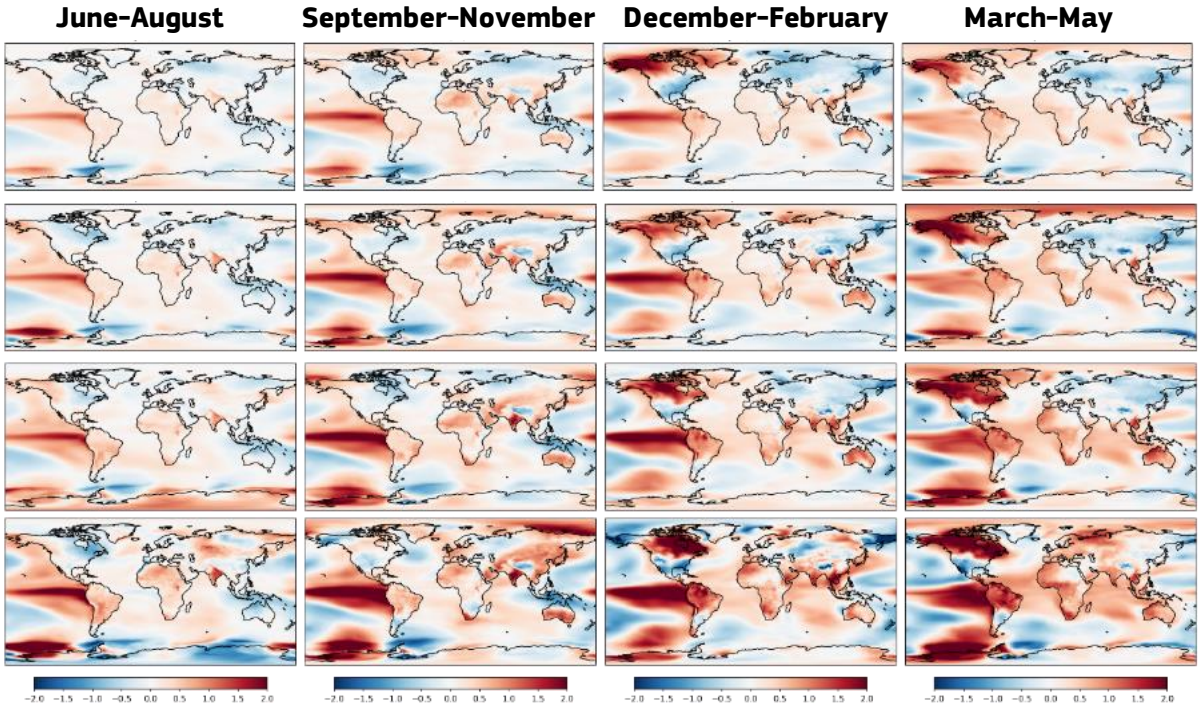
Figure 2. Classification of El Niño event intensity (top) and global surface temperature (bottom) based on the NINO3.4 intensity during Nov-Jan. Lines are colour-coded from dark red (unprecedented) to grey (neutral/La Niña), with ±1 standard deviation shading.



Source: JRC, based on the CESM2 pre-industrial model simulation of the last millennium.

Figure 2 (bottom) shows global mean surface air-temperature anomaly based on the NINO3.4 Index categories. The signal is smaller in magnitude than for NINO3.4 index (up to ~0.5 K for unprecedented events) and notably does not peak at the same time. The global temperature response lags and continues rising even after the El Niño peak. It is worth noting that stratifying the years based on their El Niño intensity also stratifies in the same order the mean global temperature indicating the central role that El NIÑO has in modulating year-to-year variability of planetary temperature.

Figure 3. Seasonal temperature anomalies of years with strong (first row from the top), very strong (second row), super (third row) and unprecedented (fourth row) El Niño intensity during Nov-Jan.



Source: JRC, based on the CESM2 pre-industrial model simulation of the last millennium.

Figure 3 displays a grid of global maps showing surface temperature anomalies across four seasons for the top four El Niño intensity categories. Rows represent increasing El Niño intensity (strong → very strong → super → unprecedented, top to bottom) and columns represent seasons (June-Aug, Sept-Nov, Dec-Feb, March-May, left to right). The colour scale runs from blue (cooler than normal) to red (warmer than normal), ranging approximately ±2 °C.

Strong El Niño (Figure 3, first row from the top). A warming is visible over the tropical Pacific in all seasons, becoming slightly more pronounced in December-February. Australia, South America, South Asia, Africa and western North America show a general warming tendency that peaks around Dec-Feb or March-May. Northern Eurasia and southern North America show mostly a cooling signal.

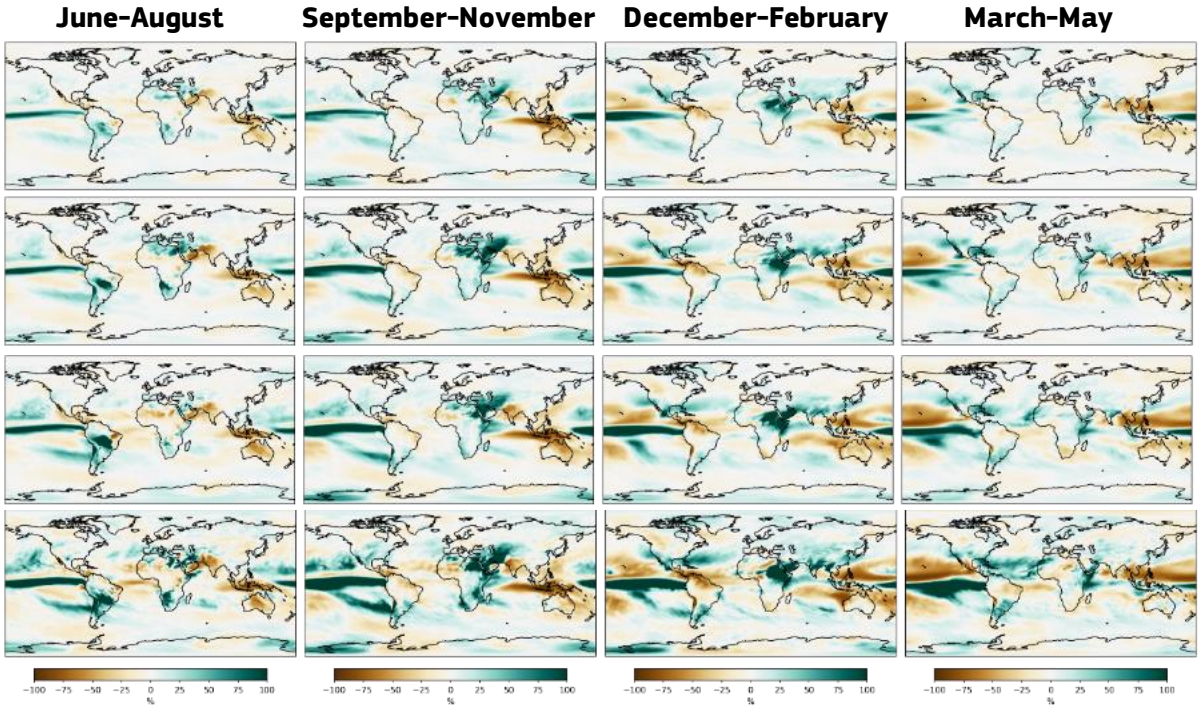
Very Strong El Niño (Figure 3, second row from the top). The tropical Pacific warming becomes more spatially extensive and intense, especially in December-February and March-May. A stronger warming band extends across the equatorial Pacific, which is precisely the ENSO region. The pattern observed for the strong El Niño maps become more accentuated except for norther Eurasia which starts showing mixed signals of warming and cooling.

Super El Niño (Figure 3, third row from the top). The equatorial Pacific warming is now very pronounced and extends well into the central Pacific. The global teleconnection pattern becomes clearer — strong warming over much of the tropics and subtropics, including Africa, South Asia, and Australia. In December-February there is a strong warm band stretching across the entire tropical belt. Colder anomalies are visible over parts of the North Atlantic and northwestern North America in some seasons, consistent with known El Niño teleconnection patterns.

Unprecedented El Niño (Figure 3, fourth row from the top). The tropical Pacific warming is at its most extreme, with warming covering the entire equatorial band. The global warming signal is the

most spatially coherent and intense of all rows. Virtually the entire tropical belt — Africa, South Asia, Southeast Asia, Australia, Central and South America — shows strong positive temperature anomalies (up to 2 °C) particularly in December-February and March-May. The warming persists and even strengthens into the post-peak season (March-May), reflecting the large thermal inertia. Some blue (colder) anomalies remain visible over the subpolar North Atlantic and parts of North America, suggesting a weakened or shifted jet stream response. In this most extreme case, a nonlinear behaviour emerges over the Euro-Asian continent with broader positive temperature anomaly response in March-May.

Figure 4. Seasonal precipitation anomalies (percent) of years with strong (first row from the top), very strong (second row), super (third row) and unprecedented (fourth row) El Niño intensity during Nov-Jan.



Source: JRC, based on the CESM2 pre-industrial control model simulation of the last millennium.

Figure 4 is as Figure 3, but for precipitation. Colour scale runs from brown (drier, up to -100%) to green (wetter, up to +100%).

Strong El Niño (Figure 4, first row from the top). Modest precipitation anomalies. A drying tendency is visible over Indonesia, Australia and in India only during June-August. The central-eastern Pacific shows some wetting. There is also a clear signal of much wetter conditions in East Africa and the Middle East between September and February.

Very Strong El Niño (Figure 4, second row from the top). The classic El Niño precipitation dipole becomes more clearly visible. Large parts of Southeast Asia and most of Australia show clear drying (brown). The central and eastern equatorial Pacific shows enhanced rainfall. The patterns over East Africa and the Middle East show increased precipitation in September-November and December-February. Southern Africa begins to show drying at the peak of the rainy season December-February. The Amazon shows some drying tendency in December-February and March-May, but central South America shows wetter conditions in June-August.

Super El Niño (Figure 4, third row from the top). The patterns intensify considerably. The drying over Indonesia and Australia is stronger and spatially coherent across all seasons. Southern Africa shows pronounced drying particularly in December-February and March-May, consistent with known El Niño drought impacts in the region. The wetting signal over the central-eastern Pacific is very strong, while Central America and the Caribbean show drying. The Indian subcontinent shows reduced monsoon rainfall in June-August, consistent with the documented ENSO-monsoon link. The La Plata basin in southeastern South America shows enhanced rainfall in December-February, a well-known El Niño signature.

Unprecedented El Niño (Figure 4, fourth row from the top). Most of the patterns described above are amplified to their maximum extent. The drying over Indonesia, Australia, southern Africa, and Central and northern South America is extreme (deep brown, approaching -75% to -100% in some areas). The wetting over the central-eastern Pacific and southeastern South America is equally intense. Notably, the Indian Ocean sector shows a complex pattern — some areas of eastern Africa remain wet while southern and southeastern Africa dry out. The June-August panel already shows strong anomalies, suggesting the impacts begin earlier in the year for unprecedented events, which is consistent with the earlier onset visible in Figure 2.

Figure 5 is as Figure 3, but for soil moisture, which in part integrates precipitation over time making the patterns smoother and more persistent than precipitation, but also integrates the impacts of changes in evapotranspiration due to temperature, radiation, and humidity anomalies. Negative soil moisture anomalies are an indicator of agricultural drought.

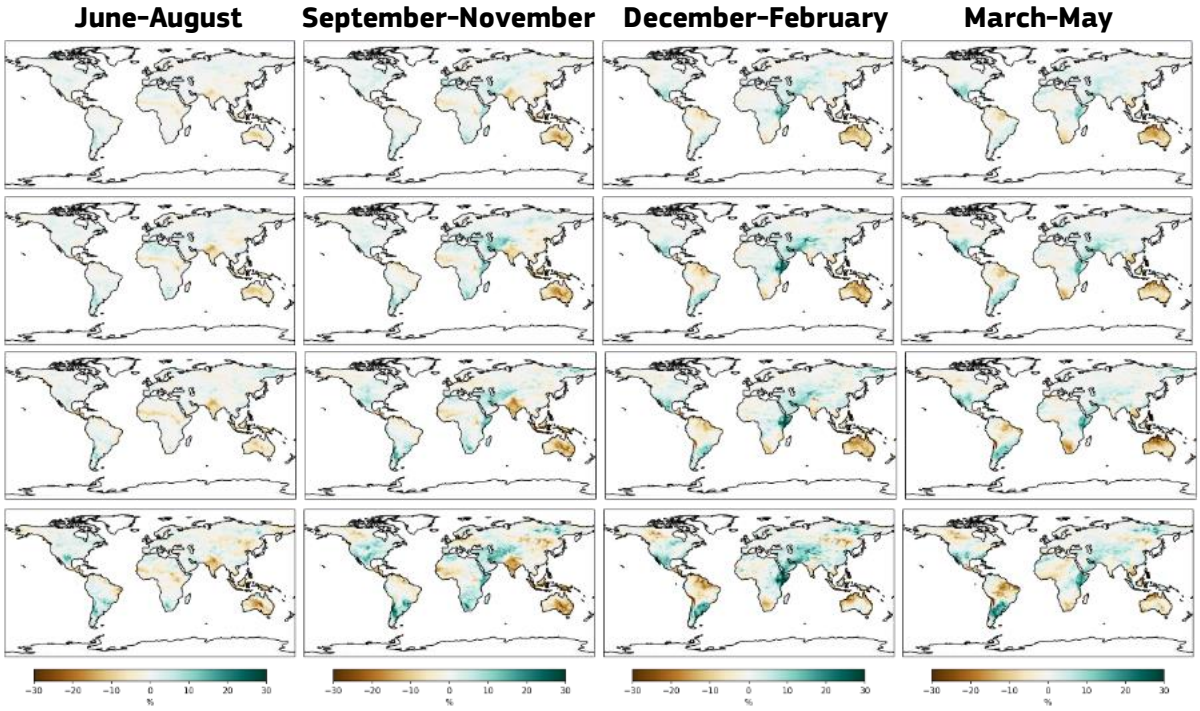
Strong El Niño (Figure 5, first row from the top). The signals are generally small. There is a consistent drying over most of Australia, southern Africa, and norther Amazon and wetter conditions in la Plata, East Africa and southern North America.

Very Strong El Niño (Figure 5, second row from the top). The patterns are very similar to the ones from strong El Niño events but more marked. Clearer drying signal emerges over northern and eastern Australia, particularly from September-November onwards. Southern Africa still shows moderate drying in December-February and March-May. Parts of Central America and northern South America also continue to show drying, while the signal over South and Southeast Asia is weak at this intensity. Dry conditions over the Sahel start to emerge more clearly during the main rainy season (Jun-Aug).

Super El Niño (Figure 5, third row from the top). The drying signal in Australia is robust across all four seasons and covers most of the continent. Southern Africa shows consistent drying in December-February and March-May. Parts of Central America and the Amazon periphery also show soil moisture deficits. Interestingly, large areas of eastern Africa, the Middle East, Central Asia, southern South America, and southern North America show wetter conditions, consistent with the precipitation signal in Figure 4. Europe shows near-neutral conditions throughout.

Unprecedented El Niño (Figure 5, fourth row from the top). The drying signal over India, southern Africa, and parts of Central and South America is at its strongest. Australia instead shows a weaker signal in unprecedented than in super El Niño events. Eastern Africa retains a strong positive soil moisture signal coherent with precipitation. Northern and northeastern Europe start to show important drying in the autumn, while the western Mediterranean shows signs of wetter conditions.

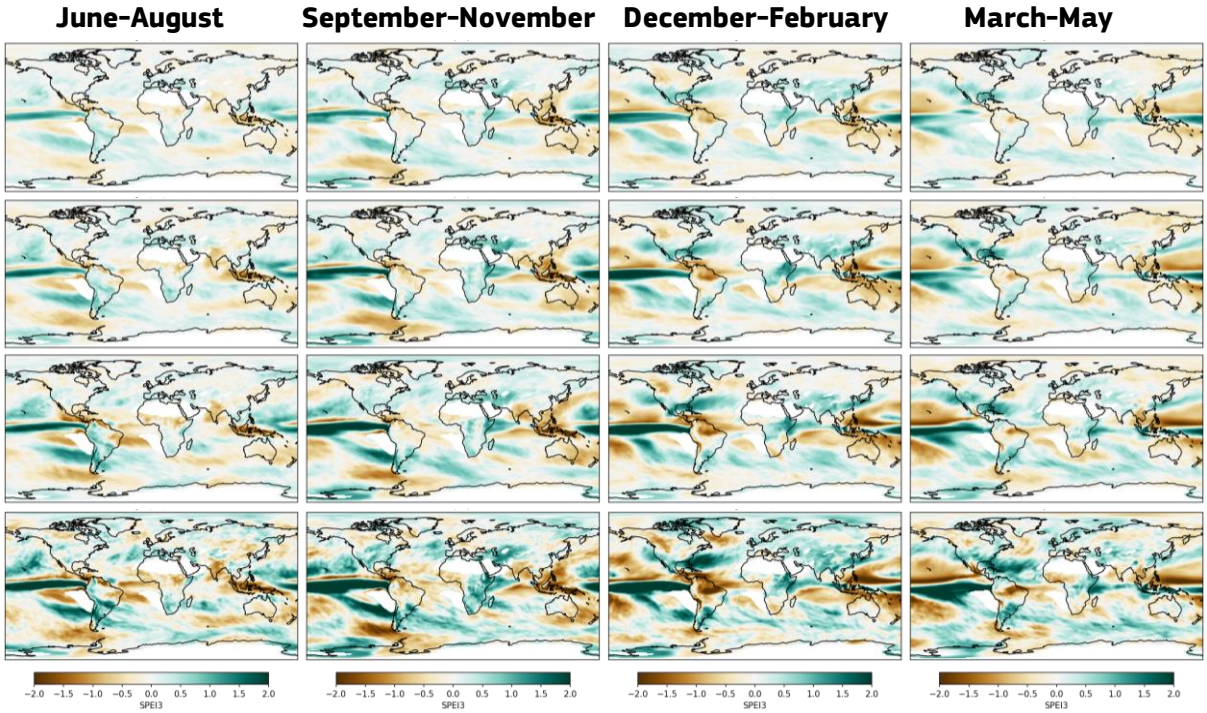
Figure 5. Seasonal soil moisture anomalies (percent) of years with strong (first row from the top), very strong (second row), super (third row) and unprecedented (fourth row) El Niño intensity during Nov-Jan.



Source: JRC, based on the CESM2 pre-industrial control model simulation of the last millennium.

Figure 6 is as Figure 3, but for the 3-month Standardised Precipitation Evapotranspiration Index (SPEI-3) which considers the combined effect of both precipitation and potential evapotranspiration. The SPEI complements all the previous indices and provides relevant information in terms of water balance. The SPEI-3 maps (Figure 6) qualitatively agree very well with the precipitation (Figure 4) and soil moisture (Figure 5) responses to El Niño. That is, the strength of the signals tends to grow with higher intensity, and the affected areas of dry and wet conditions coincide for precipitation, soil moisture and SPEI. One important thing to notice is that some areas that start to recover in terms of water balance at the end of the El Niño cycle (March-May), such as Australia, still show a strong signal in soil moisture likely due to the lagged response of the soils to water stress.

Figure 6. SPEI-3 during years with strong (first row from the top), very strong (second row), super (third row) and unprecedented (fourth row) El Niño events during Nov-Jan.

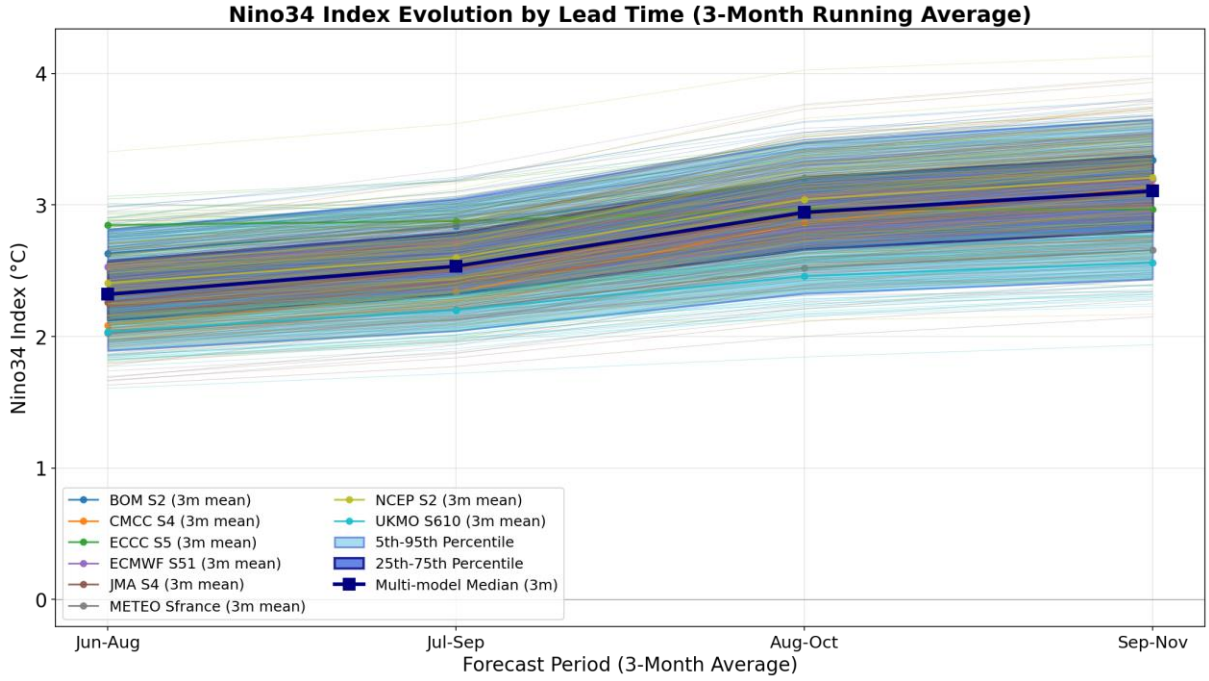


Source: JRC, based on the CESM2 pre-industrial control model simulation of the last millennium.

3. Forecasts

Figure 7 displays the temporal evolution of the NINO3.4 index based on the latest (June 2026) seasonal forecasts available in the Copernicus Climate Change Service C3S. The multi-model median is above the "very strong" limit of 2.0 °C already for the June-August period and reaches the 3.0 °C limit of "Unprecedented El Niño" by September-November. Many individual forecast systems (e.g. BOM, Bureau of Meteorology of Australia) are above the "unprecedented" category by September-November. Additionally, all analysed systems have a median larger than 2 °C by June-August highlighting the high likelihood of experiencing at least a very strong El Niño event this year (100% for August-October and September-November). Note also that the peak of El Niño activity typically occurs during November-January. The Deutscher Wetterdienst (DWD) forecasts were excluded due to potential inconsistencies in the initial phase of the simulations.

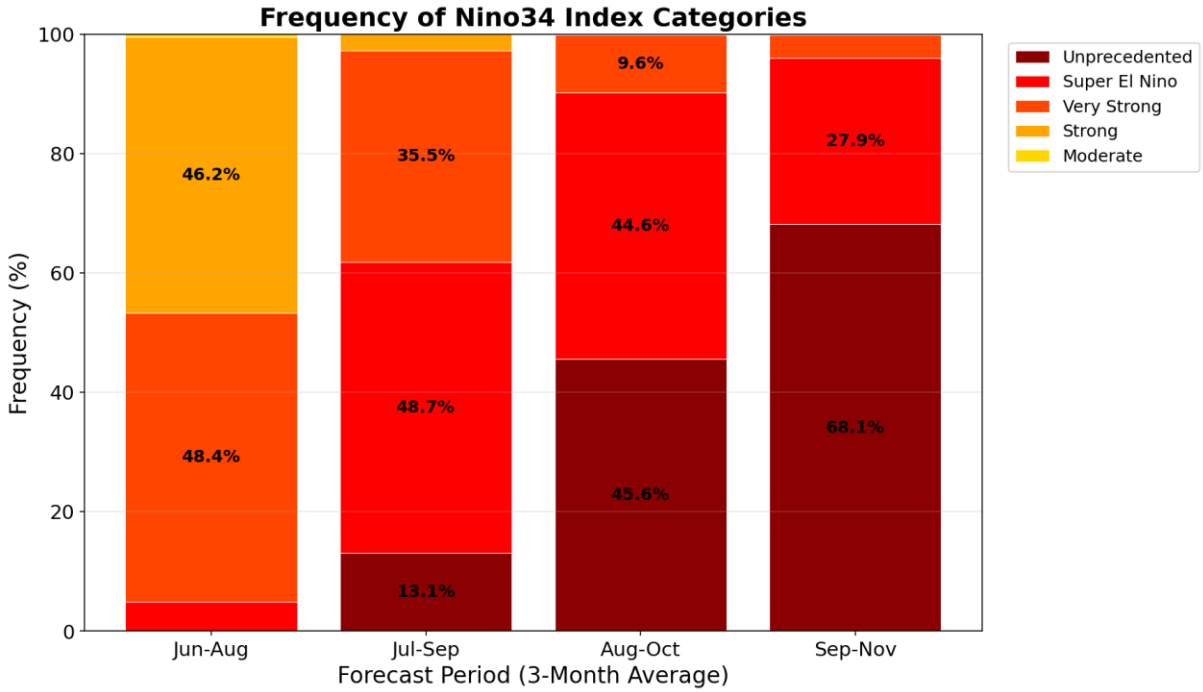
Figure 7. NINO3.4 forecasts from the Copernicus C3S multi-model ensemble. Forecasts issued in June 2026.



Source: JRC based on Copernicus C3S data.

In terms of likelihood (Figure 8), the forecast systems predict about 95% probability of a strong/very strong El Niño by June-August. As the strength of the event develops later during the year, the likelihoods in September-November become around 68% for an unprecedented event, 28% for a super El Niño and about 4% for the less extreme categories. Note that the likelihood of neutral or La Niña is 0%, indicating that the development of El Niño in the coming months is virtually certain according to these forecasts.

Figure 8. Likelihood of NINO3.4 forecasts categories from the C3S multi-model ensemble. Forecasts issued in June 2026.



Source: JRC, based on Copernicus C3S data.

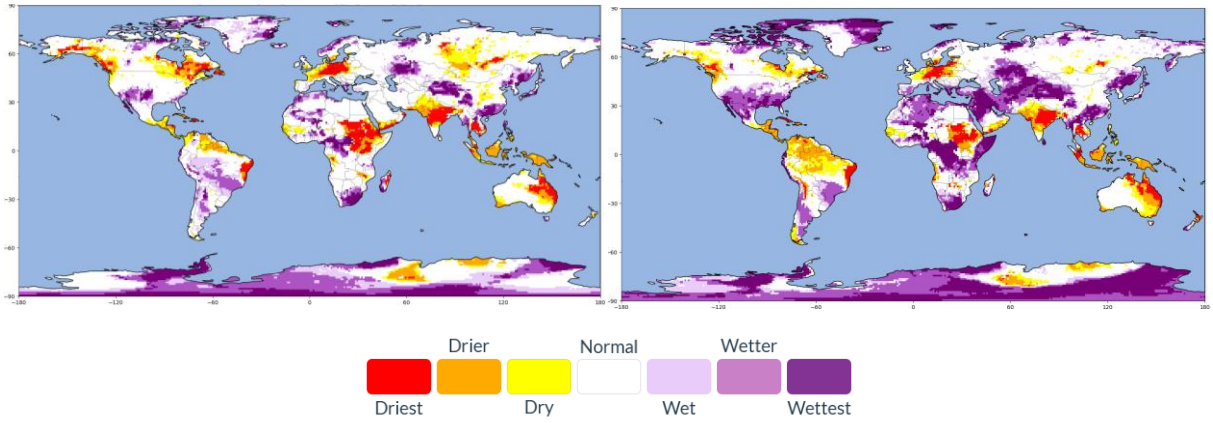
Figure 9. Forecasts of July (left panel), August (central panel) and September (right panel) of warm (red shading) and cool spell (blue shading) intensity likelihood. Forecasts issued in June 2026.



Source: JRC, Copernicus Emergency Management Service Global Drought Observatory.

Figure 9 displays the likelihood of warm and cold spells intensity during July, August and September of 2026 based on multiple seasonal forecast systems, expressed by the Multi-System Indicator for Forecasting Unusual Warm and Cool Conditions of the CEMS (Copernicus Emergency Management Service) Global Drought Observatory (GDO). There is a consistent maximum likelihood of warm spell/heatwave activity in the tropical and subtropical band in all three months, except for Australia which shows normal conditions for almost the whole period. Western North America, Europe (except the north), the Middle East and Central Asia show a consistent high likelihood of heatwave activity throughout the summer. These maps tend to qualitatively agree well with the average impact of El Niño on June-August temperature shown in Figure 3.

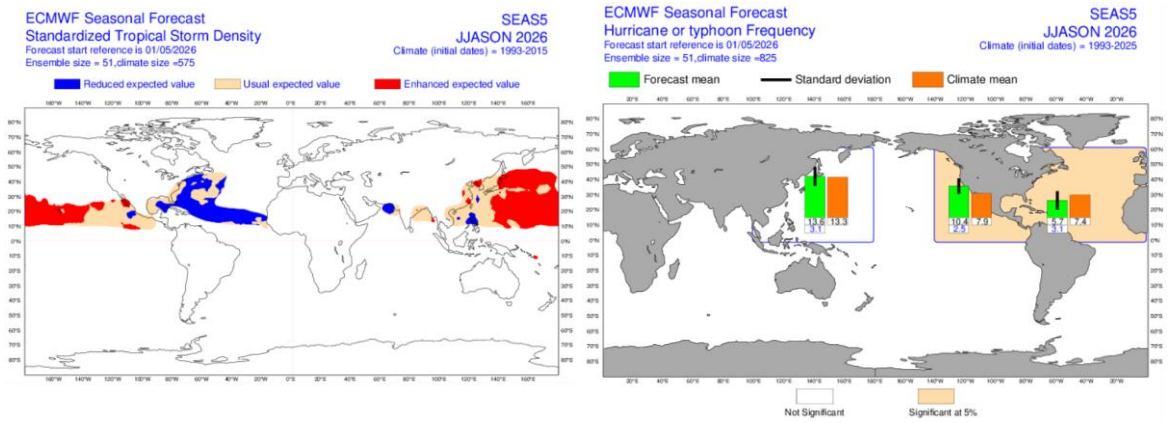
Figure 10. Dry and wet forecast likelihood for the periods June-Aug (top) and June-Nov (bottom). Forecasts issued in June 2026.



Source: JRC, Copernicus Emergency Management Service Global Drought Observatory.

Figure 10 shows the likelihood of experiencing below (yellow/red) and above (purples) precipitation for the periods June-August (left) and June-November (right), expressed by the Multi-System Indicator for Forecasting Unusually Wet and Dry Conditions of the CEMS GDO. Both maps share many similarities, such as strong signals of dry conditions in north/northeastern Europe, India, Eastern Brazil, Central America, China and Mongolia, the south-east Asia, eastern Sahel and eastern Australia, and wet conditions in southern North America, and southern Africa. Key differences appear in the Amazon that turns from wet to dry conditions, the Middle East, the extended Horn of Africa, central Africa and Central Asia turn from near normal to wet. The forecasts presented in Figure 10 tend to agree qualitatively well with the average impacts expected from El Niño on precipitation (Section 2).

Figure 11. June-November forecast of the likelihood in changes of areas of development of tropical cyclones (left) and the total expected number of hurricane and typhoon formation (right).



Source: ECMWF.

Figure 11 shows the ECMWF (European Centre for Medium-Range Weather Forecasts) seasonal forecasts of standardised tropical storm density anomalies for June–November 2026 on the left map. Blue areas indicate reduced expected storm activity; red areas indicate enhanced activity. The main signals are suppressed activity in the Atlantic basin (a classic El Niño signature) and enhanced activity in parts of the western Pacific and Indian Ocean. The right map shows the hurricane and typhoon expected numbers, with box plots overlaid on key ocean basins showing forecast mean,

standard deviation, and climate mean. The Atlantic shows below-normal activity while the western Pacific and Indian Ocean basins show near-normal to slightly above-normal values, however only the eastern Pacific and Atlantic show statistically significant deviations from the long-term average. These forecasts from ECMWF are very much in line with the latest NOAA estimates which give a 5% chance of lower Atlantic hurricane activity than usual and a 70% probability of higher activity in the Pacific (Gardner, 2026). Preparedness is still essential, despite the lower number of storms expected in the Atlantic, the higher temperatures of the waters could make the storms stronger.

The GloFAS (Global Flood Awareness System) seasonal discharge forecast, initialised in June 2026, indicates predominantly below-normal to low discharge outlook across several major river basins globally, with signal strength and spatial coherence decreasing progressively at longer lead times as forecast uncertainty grows.

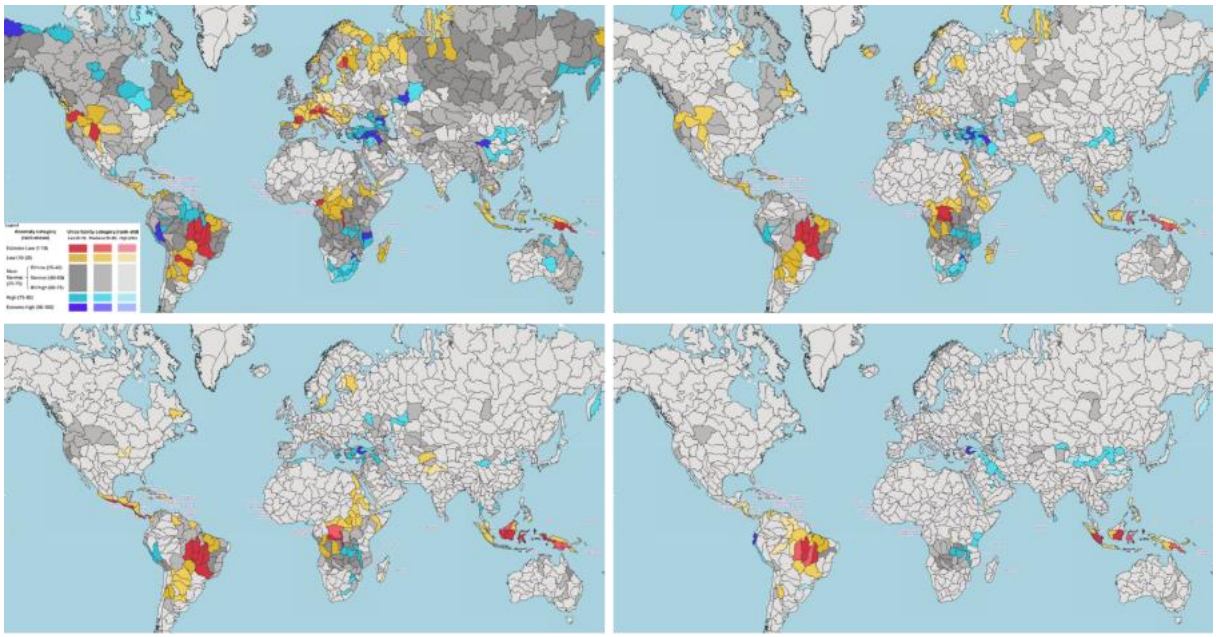
At the 1-month lead time (Figure 12, upper left), the forecast exhibits the strongest and most spatially coherent anomaly signal. Below-normal to extreme low discharge conditions (red and orange) dominate across central and eastern Europe including the Baltic Sea region, some basins in the western/central US, the Congo basin in central Africa, parts of southern and southeastern Asia with a hotspot (having high confidence) in Indonesia and Papa New Guinea. Many basins across South America that differ from normal conditions show low to extreme low flow anomalies for the 1-month lead time. In contrast, some basins in the eastern Mediterranean, south and southeastern Africa, eastern Asia, North America as well as the northern part of South America show positive flow anomalies. These signals carry predominantly low-to-medium uncertainty.

At 2-month lead time (Figure 12, upper right), the spatial pattern is partly broadly maintained but with increasing uncertainty and a gradual transition of many basins toward near-normal conditions (grey). The below-normal signal over central Africa and large parts of South America together with the wetter conditions in south and southeastern Africa persists with moderate to high confidence, while the above-normal signal over parts in the Amazon and North America begins to weaken and become more spatially fragmented. While Central Europe turns more towards normal conditions, the dry anomalies move further northeast towards single basins in Russia and some smaller parts around the Baltic Sea. Significant above-normal conditions (low uncertainty) are furthermore observed in Türkiye and the Middle East.

By 3-month lead time (Figure 12, lower left), the forecast signal of either wet or dry conditions is more diffuse globally accompanied by 3 more distinct and persistent regions. The largest portions of Europe, North America and Asia transition to near-normal conditions, although being subject to high uncertainties. In contrast central Africa is one of the more distinct regions that retains a large fraction with below-normal flow signals. The same counts for Central America as well as central and northeastern South America where substantial parts show low to extremely low conditions even with high confidence. Analogously, Papa New Guinea and Indonesia show low to extremely below normal conditions with high confidence. Fragmented parts of southeastern Africa remain wetter than normal although overall the impacted area slightly decreases, and the level of certainty of the predictions is mostly low at this stage.

At 6-month lead time (Figure 12, lower right), the global picture is largely near-normal with high uncertainty across most basins, as expected at this extended forecasting range. Residual below-normal signals persist over parts of Indonesia, central to northeastern South America. While scattered above-normal signals appear in parts of South America's west coast, in east/southeast Africa, Türkiye and Middle East, and Central China.

Figure 12. GloFAS seasonal river discharge forecast for global river basins initialised in June 2026. Each panel shows the monthly mean discharge anomaly on the basin scale expressed as a rank percentile relative to the 20-year range-dependent climatology (2004-2023), for lead times of 1 month (upper left), 2 months (upper right), 3 months (lower left), and 6 months (lower right). Colours indicate the anomaly category (rank-mean) ranging from Extreme Low (1–10th percentile, dark red) to Extreme High (90–100th percentile, dark blue/purple), while colour intensity reflects the uncertainty level (rank-std): dark shades indicate low uncertainty (0–10), medium shades medium uncertainty (10–20), and light shades high uncertainty (>20). Basin values are computed as a weighted average (by the square root of catchment area) of the anomaly and uncertainty signal at river pixels with upstream area exceeding 1000 km^2. Grey shading indicates near-normal conditions (25–75th percentile range).



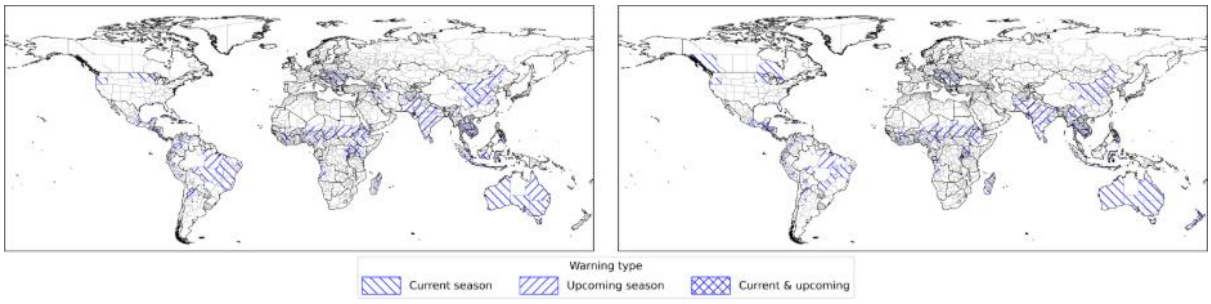
Source: JRC, Copernicus Emergency Management Service GloFAS.

4. Crops and food prices

Agriculture represents one of the main transmission channels of socio-economic impacts driven by the climate variability and extremes caused by El Niño. Droughts, floods and temperature anomalies can disrupt crop production and global food trade, with effects propagating through interconnected food systems and commodity markets (Adolfsen and Lappe, 2023). The resulting impacts on food security, household purchasing power and public finance are particularly relevant for vulnerable low- and middle-income countries, especially where El Niño shocks interact with existing socio-economic fragilities. Given these effects, we look at cropland and pasture and analyse the likely impacts on prices of primary staples. The literature investigating the relationship between ENSO events and crop yields (Iizumi et al., 2014; Legler et al., 1999), and agricultural commodity prices (Ubilaya, 2018; Peri, 2017) is extensive, and generally confirms a significant effect of this phenomenon on both outcomes. However, uncertainties and model-related limitations (as well as assumptions) remain.

Figure 13 shows the warnings released by the JRC Anomaly Hotspots of Agricultural Production (ASAP) system and based on seasonal precipitation forecasts and specifically tailored for cropland and pastoral areas. Precipitation forecasts are based on the Copernicus C3S multi-model ensemble of seasonal forecasts issued in May. Agricultural areas with negative prospects are aligned with the scenarios depicted in Figure 5, 6, and 10, and include areas in Sub-Saharan Africa, India, China, Australia, Brazil, and Eastern Europe. Southern Africa is not affected as the current growing cycle is coming to an end and the next one starts beyond the forecast horizon (6 months, up to October), but impact is likely in case the El Niño anomaly will propagate into 2027.

Figure 13. JRC-ASAP precipitation forecast warnings issued on 11 May 2026 for cropland (left) and pastoral areas (right). Warnings are represented by blue hatching (backward and forward hatching denote a warning for the remaining part of the current growing season and the next growing season, respectively). Warnings denote administrative units where the most likely category is drier than usual and the forecast has positive skill.



Source: JRC.

To analyse the impacts of El Niño on global food prices, we here selected four crops considering their predominance as global staples: maize, rice, soybeans and wheat. The last one is further divided into Hard Red Winter (HRW) and durum. This analysis aims at providing scenario-based predictions of price evolution of the selected commodities over the same period covered by the current ENSO forecasts. A stacked ensemble model is adopted, combining traditional econometric models with machine learning (ML) algorithms. Five models – ARDL (Autoregressive Distributed Lag), SARIMAX (Seasonal Autoregressive Integrated Moving Average with Exogenous Variables), Elastic Net, Random Forest and XGBoost – have been trained on historical series of commodity prices and the NINO3.4 index. The predictions generated by these models on past prices are then

used as explanatory variables by a "meta" model – specifically a ridge regression – to produce the final price forecast.

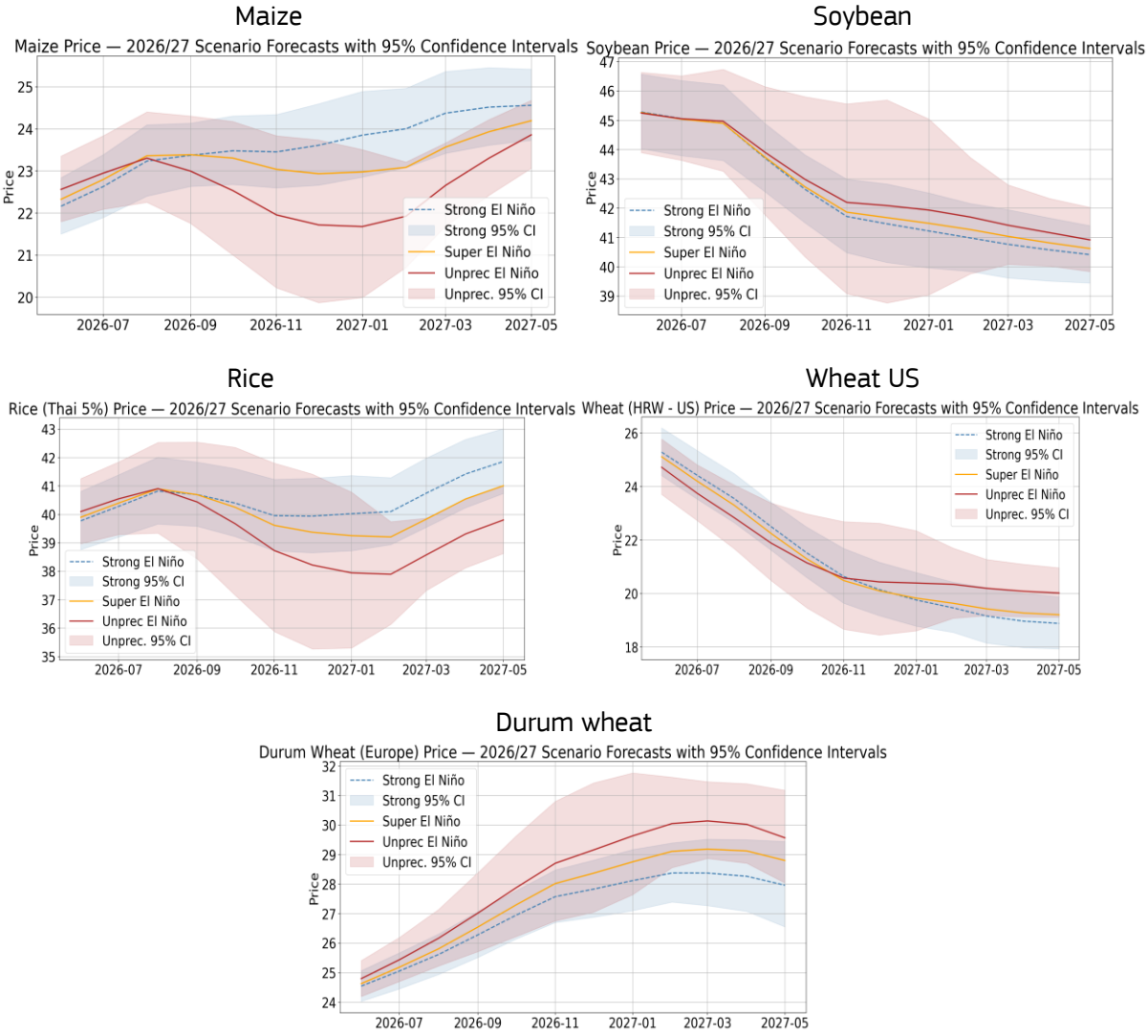
Price series have been obtained from the World Bank "Pink Sheet"[4], as monthly nominal dollar values, with the exception of durum wheat, whose price is sourced from the European Commission Agri-food Data Portal[5] and is expressed in nominal euros. The World Bank price series are derived as averages of sale transactions at specific representative ports – the US Gulf for maize and HRW wheat, Bangkok for Thai 5% rice, and the US Gulf/Rotterdam for soybeans – whereas the durum wheat price represents an average across European member states. The models have been trained on data from January 1960 to March 2026, except for durum wheat, whose price series begins in January 1991.

Figure 14 shows that the overall evolution of the price series is broadly similar across all considered ENSO scenarios. However, caution should be used when interpreting super and unprecedented scenarios given the associated potential nonlinear response which cannot be taken into account by the trained models (especially for the unprecedented category). Among the commodities examined, soybeans and HRW wheat display a clear downward trend over the forecast horizon. According to Iizumi et al. (2014), soybeans are among the crops most positively affected by El Niño globally, with average yield gains of 2.1–5.4%. Concerning HRW wheat the detrimental effects on US wheat production reported in some studies (Ubilava, 2018; Anderson et al., 2017; Iizumi et al., 2014) may become apparent after June 2027 – beyond the current forecast horizon. At the opposite end, durum wheat shows a clearly upward trend. While global maize prices show a modest increase during summer 2026, together with an initial decrease for super and unprecedented categories. The last pattern, as already highlighted, should be interpreted with caution given that the unprecedented El Niño did not enter into the training and due to potential nonlinear responses as shown by Figure 3. Rice prices, overall, tend to remain stable/ slightly decrease price and have a tendency to increase in mid-2027. The reference price for rice here used is the Bangkok market one, which reflects Thai and Southeast Asian supply more than South Asian supply. Thailand and Vietnam actually tend to benefit from El Niño in certain phases – drier conditions reduce flooding in the Chao Phraya delta and can favor the dry-season irrigated crop. Although South Asian monsoon failures do reduce yields in India and Bangladesh, India's export restrictions mean that this supply shock does not fully transmit to the Bangkok price (Limsakul, 2019).

[4] https://www.worldbank.org/en/research/commodity-markets (5 April 2026, last access: 29 May 2026)

[5] https://agridata.ec.europa.eu/extensions/DataPortal/prices.html (Last access: 29 May 2026)

Figure 14. Predicted price ($/100Kg) evolution of selected food commodities from June 2026 to May 2027 under strong, super and unprecedented El Niño scenarios. The shaded areas represent the confidence interval with 95% confidence.



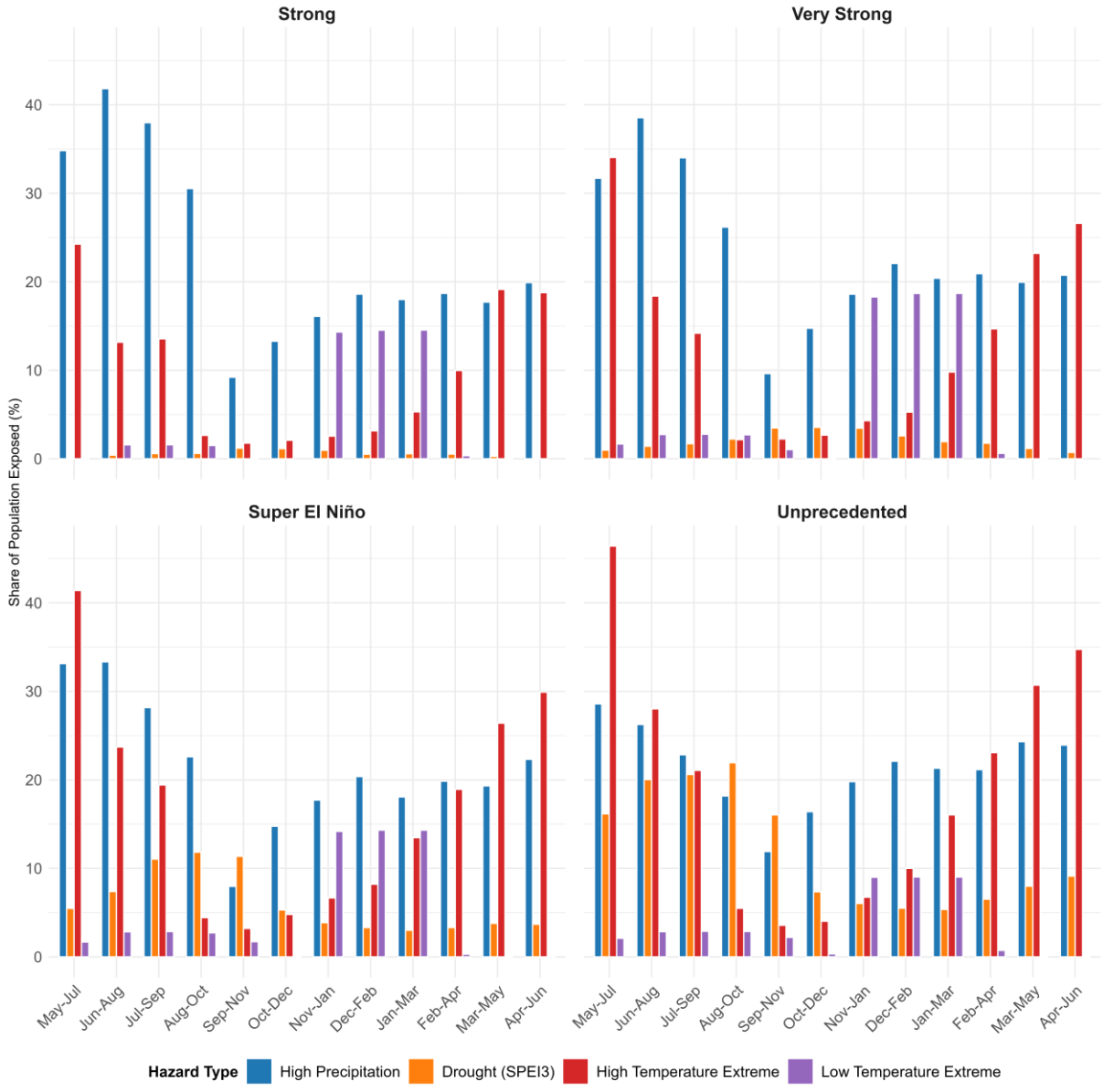
Source: JRC.

In summary, the coming El Niño phase is very likely to impact global agricultural prices, particularly if the most extreme scenarios will materialize. However, uncertainties emerge especially for some crops. The present global price analysis is based on a small sample of crops whose prices should not be considered a perfect representation of global ones. Despite being taken as reference by the World Bank, such prices may be strongly determined by local market conditions (e.g. the U.S. market for maize and wheat HRW, and Southeast Asia for rice). Given the diverse impact of El Niño in different world regions, and the far from perfect price transmission from international to local markets (Emediegwu and Rogna, 2024), a thorough analysis of food security should be conducted on a regional/local level. In addition to major crops, the potential impacts of El Niño on subsistence agriculture (i.e crops and livestock) and dependent livelihoods (about two billion people), may be significantly larger.

5. Population exposure and humanitarian risks

The analysis in this section examines potential population exposure to four key hazards (above-normal precipitation, drought, high temperature extremes, and low temperature extremes) across the different El Niño categories. By using global population and hazard exposure data, we quantify the share of population exposed to each hazard type, revealing spatial patterns and vulnerability hotspots.

Figure 15. Share of global population exposed in each period according to the different El-Niño categories and hazard type. Hazards are defined as follows. High precipitation: 3-month accumulated precipitation exceeds its 84th percentile. Drought: SPEI-3 is below –1. High temperature extreme: 3-month mean surface air temperature exceeds its 95th percentile. Low temperature extreme: 3-month mean surface air temperature is below its 5th percentile.



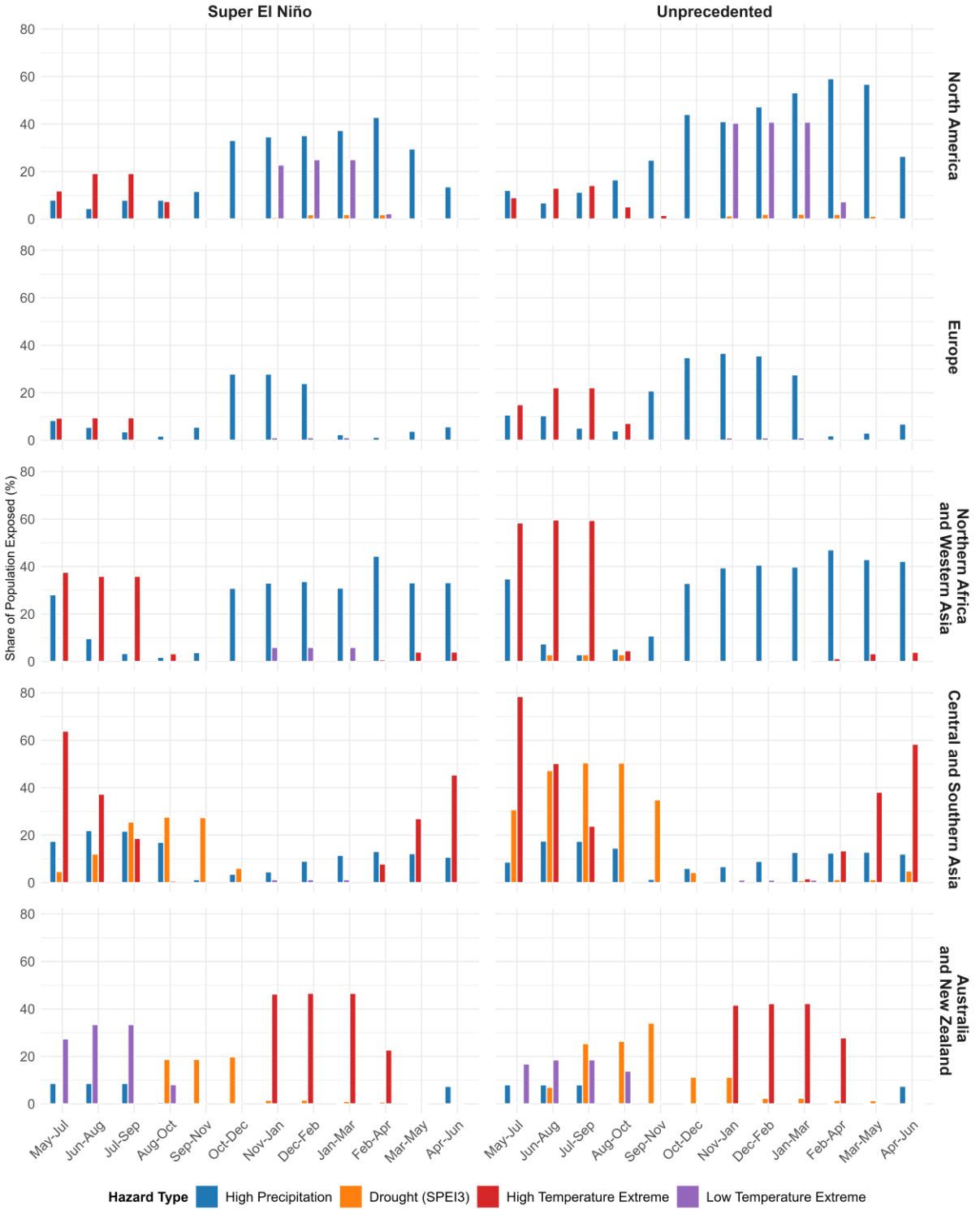
Source: JRC based on GHS-POP R2023A for year 2025.

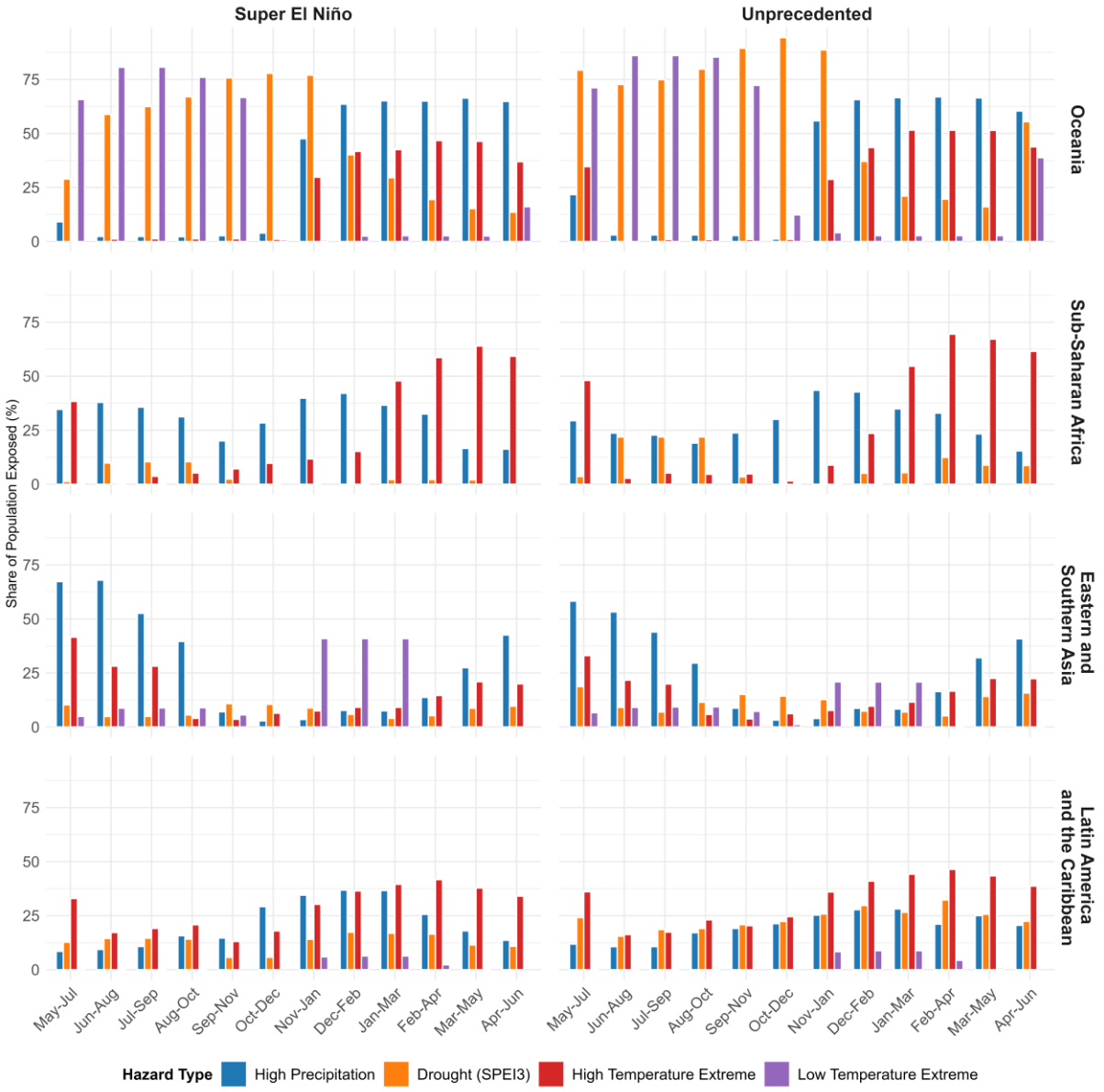
The data in Figure 15 clearly shows that overall population exposure to climate hazards increases with stronger El Niño events (from strong to unprecedented). However, this increase is not uniform and is more intense in specific regions. Globally, the overall exposure reaches shares above 20% for low temperature extremes, all the while the share of population exposed to droughts and high temperature extremes also rises considerably.

All regions demonstrate distinct seasonal patterns in hazard exposure. Regions in the Northern Hemisphere may experience summer heat extremes (June-August) and winter precipitation extremes (December-February). Regions in the Southern Hemisphere show summer heat extremes (December-February) and winter precipitation ones (June-August). Tropical regions exhibit monsoon-related patterns with drought and precipitation extremes that can create complex, compound hazard scenarios. Sub-Saharan Africa may experience more or longer high temperature extremes, while Eastern and Southeastern Asia will be exposed to intensification of both drought and high temperature extremes. Latin America will face increased exposure across multiple hazards during stronger El Niño phases. These patterns underscore the need for progressive scaling of preparedness efforts as El Niño intensity increases.

Importantly, the stronger the El Niño, the more prominent is population exposure to high temperature and droughts. This feature has very relevant policy implications not only for food safety implications, but also for health-related impacts on the population in affected regions.

Figure 16. Exposure by hazard and stronger El Niño events categories for selected UN SDG Regions.





Source: JRC based on GHS-POP R2023A for year 2025.

Oceania may face the highest overall exposure to multiple hazards, especially in the two worst-case scenarios: super and unprecedented. Albeit having a relatively scattered and small population, Oceania may be substantially impacted. Moreover, significant demographic challenges amplify the impacts of El Niño. While these countries have relatively strong infrastructure compared to many developing regions, their geographic isolation, and dependence on climate-sensitive industries may contribute to the vulnerability to climate hazards. These factors, combined with environmental pressures, create a complex landscape of risks that may require targeted policy interventions and adaptive strategies.

Indigenous communities in the Pacific Islands will also be disproportionately affected by El Niño events due to their dependence on traditional livelihoods, such as fishing and farming. During the 2015-2016 El Niño, Indigenous communities faced severe water shortages, which exacerbated health and food security issues.

Urbanisation and coastal population density presents another set of challenges. In the Pacific Islands, 40% of urban populations live in informal settlements that are highly vulnerable to flooding

and cyclones. Cyclone Winston in 2016, linked to El Niño, displaced 62,000 people in Fiji, many of whom lived in informal coastal settlements.

Sub-Saharan Africa shows high overall exposure to high temperature extremes and above-normal precipitation, which can lead to high susceptibility to heat stress and water excess. Sub-Saharan Africa faces severe demographic challenges that exacerbate the impacts of El Niño events, making the region particularly vulnerable to climate hazards. These challenges stem from rapid population growth, high poverty rates, and the vulnerability of key groups, all of which amplify the adverse effects of droughts, heatwaves, landslides and floods. Sub-Saharan Africa has the world's highest natural population growth resulting in a young population, with up to 60% of the population under the age of 25. Such demographics increase the demand for education, jobs, food, and services, straining already limited resources during El Niño-related crises. The resulting pressure on water, food, and healthcare systems makes it harder for governments to respond effectively to El Niño - induced hazards. Urbanisation adds another layer of complexity. 40% of the population lives in cities, often in informal settlements that are highly vulnerable to floods, landslides, and heatwaves. For example, Lagos, in Nigeria, faces annual flooding that displaces thousands of people, a situation that is worsened by El Niño-related heavy rains. These informal settlements lack proper drainage systems, clean water access, and resilient housing, making their inhabitants particularly susceptible to the impacts of extreme weather events and vector borne and zoonotic diseases. Undernutrition is also a significant concern, affecting 30% of children under 5 during droughts. The 2016 El Niño led to severe acute malnutrition in 1 million children across Eastern and Southern Africa. Malnourished children are more susceptible to diseases and have lower survival rates during crises, creating a cycle of poverty and poor health. Women and girls in Sub-Saharan Africa bear a disproportionate burden during El Niño events. They are responsible for 70% of water and fuel collection, tasks that become significantly harder during droughts when water sources dry up and longer distances are to be walked to obtain water or wood. This increased workload often leads to higher school dropout rates among girls, as they are required to help with household chores during crises. For example, in Ethiopia, droughts increased girls' dropout rates by 20% due to the added burden of water fetching. This not only affects their education but also perpetuates gender inequalities and limits future opportunities for women and girls.

Eastern and Southeastern Asia face severe challenges during El Niño events, as the region's high population density, rapid urbanisation, and economic dependence on agriculture amplify vulnerabilities to climate hazards. Its exposure rate is smaller with respect to other regions, however with a total population of over 2.3 billion people, El Niño events can expose up to 500 million people in case of a 20% heat-wave exposure. With over 650 million people living in rural areas and relying on rain-fed agriculture, droughts and high temperature extremes—affecting between 20 and 40% of the population during strong events—disrupt livelihoods, food security, and public health. Urban areas, home to over 50% of the region's population, experience heat stress, water shortages, and increased migration from rural to urban centres as people flee drought-affected areas. Children, the elderly, and low-income communities are particularly vulnerable, facing higher risks of malnutrition, heat-related illnesses, and waterborne diseases during extreme weather events.

The demographic pressure in Eastern and Southeastern Asia is further intensified by seasonal labour migration and agricultural dependency, which are disrupted during El Niño-related hazards. During Jul-Sep and Feb-Apr, when drought and heatwaves peak, rural communities, comprising over 60% of the population in countries like Indonesia, Vietnam, and the Philippines, face reduced agricultural productivity, leading to food insecurity and income loss. This often forces temporary or permanent migration to urban areas, increasing pressure on already strained infrastructure and

services. Additionally, heatwaves, which can affect up to 100% of the population in some months, disproportionately impact the elderly and children, leading to increased hospitalisations.

Latin America and the Caribbean face significant challenges during El Niño events, as the region's high urbanisation rates, economic dependence on agriculture, and vulnerable coastal communities amplify risks from weather hazards. With over 60% of the population living in urban areas, many of which are informal settlements, heatwaves, landslides and flooding, affecting 30-50% of the population during strong events, exacerbate health risks and infrastructure strain. Rural communities, comprising 40% of the population and heavily reliant on rain-fed agriculture, face severe drought and crop failures, leading to food insecurity and migration to cities. Children, the elderly, and low-income groups are particularly vulnerable, facing higher risks of malnutrition, heat-related illnesses, and waterborne diseases during extreme weather events. The demographic pressure in Latin America and the Caribbean is further intensified by seasonal labour migration and agricultural dependency, which are disrupted by El Niño-related hazards. During February-April and September-November, when high temperature extremes and drought peak, rural communities, particularly in Central America and the Andes, experience reduced agricultural productivity, leading to food insecurity and income loss. This often forces temporary or permanent migration to urban areas, increasing pressure on already strained infrastructure and services. For instance, during the 2018-2019 drought, over 1.4 million people in Guatemala, Honduras, and El Salvador migrated due to crop failures, contributing to overcrowding and increased informal settlement growth in cities. Additionally, heatwaves, which can affect up to 50% of the population, disproportionately impact outdoor workers, the elderly, and children, leading to increased hospitalisations.

The INFORM (Index for Risk Management) Warning[6] combines forecasts of hazards, conflicts, food insecurity, displacement and macroeconomic conditions with structural risk and crisis severity indicators to identify countries where humanitarian crisis conditions may deteriorate over the coming 12 months. It offers a three-tier analytical structure - from forecasts to plausible signals and signal aggregation, to integrated warnings combining current and structural risks - which can be tailored to a specific context, here the anticipated humanitarian impacts of El Niño over the next six months.

In the context of INFORM Warning, the potential El Niño impacts can be traced through three distinct indicators, namely dry and wet conditions, and river flow anomaly. Dry and wet condition signals are derived from seasonal predictions of the Standardized Precipitation Index SPI (Acosta Navarro et al., 2025) and country level drought and flood risk indicators, weighted by population and agricultural exposure. The river flow anomaly is derived from GloFAS seasonal outlook normalised at country level. Such forecasts are then combined with conflict, food insecurity, displacement and macroeconomic indicators to create a compound measure of dynamic risk (Warning Signal ranging between 0 and 10) for each country. The warning signals are then combined with structural risk (INFORM Risk[7]) and the severity of ongoing crises (INFORM Severity[8]) into a Warning Level score ranging from 0 to 10, where 0 indicates no risk and 10 indicates very high risk, to support crisis anticipation operations.

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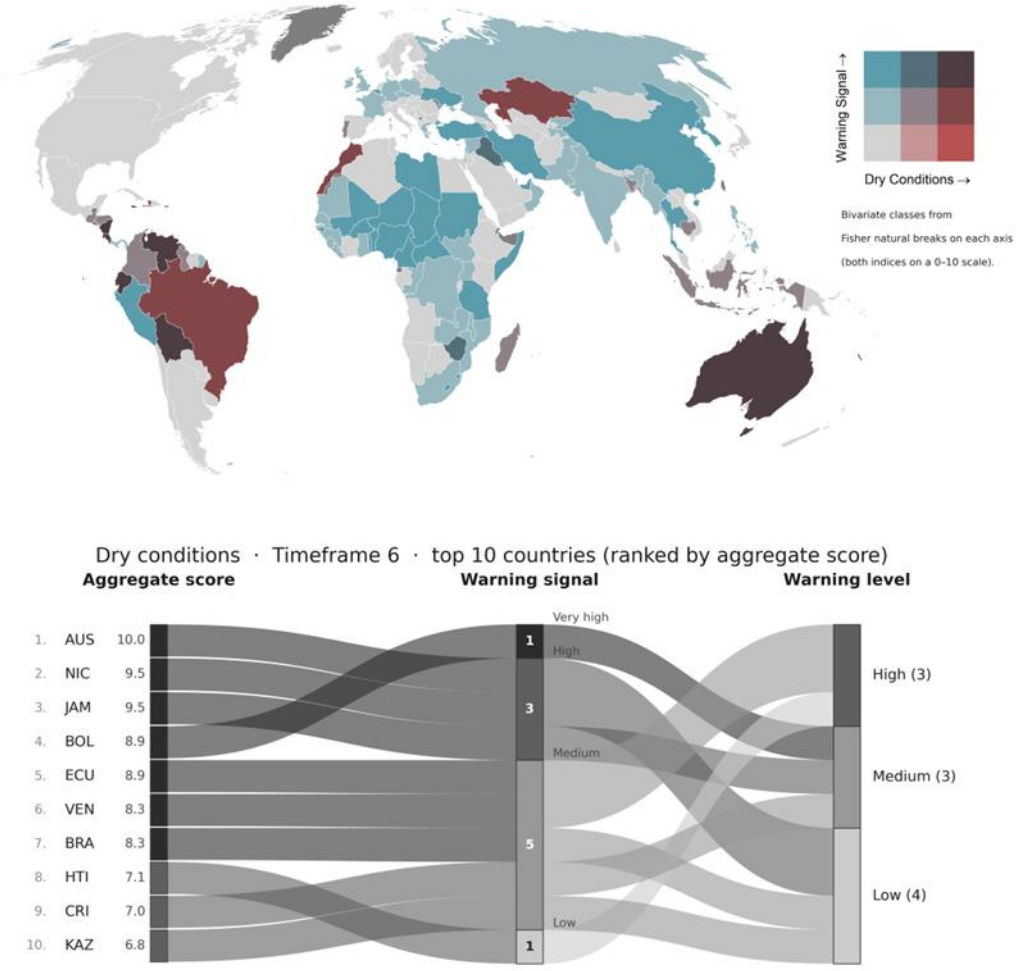
6 drmkc.jrc.ec.europa.eu/inform-index/inform-warning-tool/

7 drmkc.jrc.ec.europa.eu/inform-index/INFORM-Risk

8 drmkc.jrc.ec.europa.eu/inform-index/INFORM-Severity

Figure 17 shows the global distribution of projected dry conditions over the next six months, combined with the corresponding Warning Signal in a bivariate map, alongside a ranking of the ten countries at risk. The highest-concern areas are concentrated in northern South America, the Caribbean and southern Central America, with additional hotspots in Australia, Kazakhstan, north-western Africa, and South-East Asia. These are areas where strong dry conditions coincide with other risk factors captured in the warning signal. For example, Bolivia has the highest Warning Signal, driven by compounding effects of severe dry conditions and very strong annual inflation. Among the top ten countries with the strongest dry conditions signals, only three – Ecuador, Venezuela and Haiti – also appear on the humanitarian assistance radar through high Warning Levels. In Ecuador, INFORM Warning identifies a strong compound signal combining dry conditions, annual inflation and violence against civilians, alongside high structural risk and the severity of an existing crisis linked to conflict, violence and international displacement. In Venezuela, predicted dry conditions are compound by internal displacement and annual inflation signals over the next six months, in a country already affected by political, economic and conflict-related crises.

Figure 17. Top: a map showing co-occurrence of dry conditions and overall Warning Signal Score. Bottom: Sankey diagram of the ten countries[9] with the highest projected dry condition signal over the next six months, showing their relative ranking in terms of Warning Signal and Warning Level. All scores range from 0 to 10, with the following grouping: Very Low = 0-2, Low = 2-4, Medium = 4-6, High = 6-8 and Very High = 8-10.



Source: JRC.

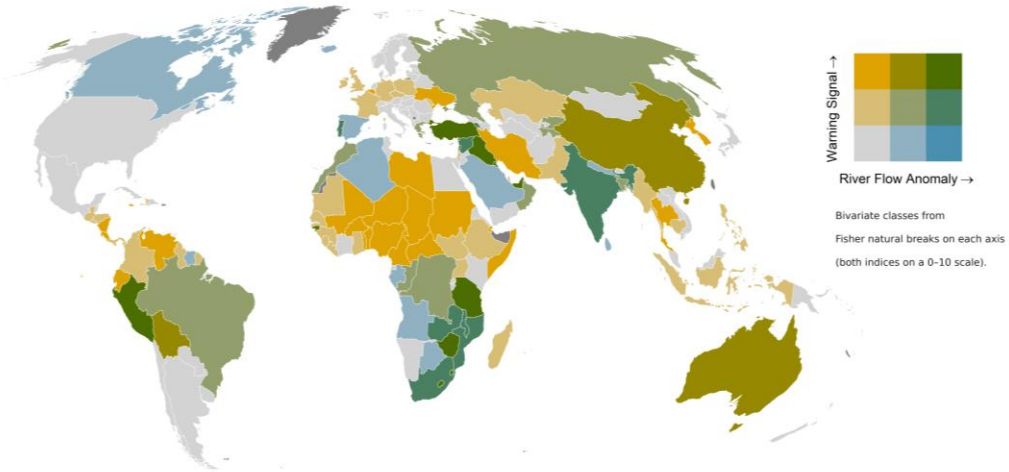
As for the wet conditions side (Figure 18), the strongest six-month population weighted SPI signals are concentrated in the region north and east of Gulf of Guinea (particularly Togo, Nigeria, Benin, Cameroon and Central African Republic). Nigeria and Central African Republic are also among the countries with the highest Warning Levels globally; forecasted wet conditions may overlap with conflict, displacement, food insecurity and economic stress, further compounding already severe crisis conditions in these countries.

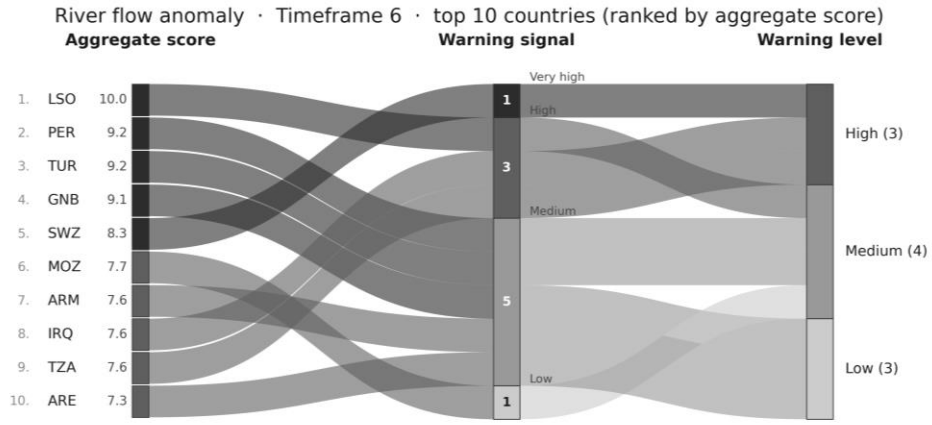
Countries with a very high river-flow anomaly signal over the next six months are mainly located in Southern Africa and Western Asia and western South America. Among these, Iraq, Eswatini, Mozambique and Tanzania show both high Warning Signal and Warning Level, reflecting the combination of forecasted hydrological anomalies with existing structural risk and crisis severity.

[9] Adopted country codes are: AUS = Australia, BOL = Bolivia, BRA = Brazil, CRI = Costa Rica, ECU = Ecuador, HTI = Haiti, JAM = Jamaica, KAZ = Kazakhstan, NIC = Nicaragua, VEN = Venezuela

Eswatini carries the largest compound Warning Signal over the coming six months, driven primarily by high river flow anomaly. In Tanzania, a strong signal for violence against civilians reinforces the elevated river flow anomaly, together producing an overall high Warning Level. Iraq presents a similar hydrological concern, where the river flow anomaly is further compounded by strong inflation and economic contraction — suggesting a context where anticipatory action could be considered over the next six months.

Figure 18. Top: a map showing co-occurrence of high river flow anomaly as deduced from GloFAS and overall Warning Signal Score. Bottom: Sankey diagram of the ten countries[10] with the highest projected High River Flow Anomaly signal over the next six months, showing their relative ranking in terms of Warning Signal and Warning Level. All scores range from 0 to 10, with the following grouping: Very Low = 0-2, Low = 2-4, Medium = 4-6, High = 6-8 and Very High = 8-10.





Source: JRC.

[10] Adopted country codes are: ARE = the United Arab Emirates, ARM = Armenia, GNB = Guinea-Bissau, IRQ = Iraq, LSO = Lesotho, MOZ = Mozambique, PER = Peru, SWZ = Eswatini, TUR = Türkiye, TZA = Tanzania

6. Human mobility

El Niño affects large parts of the world simultaneously and modify hazards' global patterns. These effects may translate into decisions to move, to be forced to move, or being unable to move. Climate hazards rarely act alone: they interact with existing vulnerabilities, economic conditions, governance and the social ties that shape where people go and whether they move at all. The impacts are not uniform: the same hazard can produce mass displacement in one context and barely measurable movement in another, depending on who is affected and what resources they must respond.

Climate hazards fall into two broad categories with distinct mobility dynamics. Sudden-onset events, such as floods and storms, often generate immediate, visible displacement. People evacuate rapidly, often to nearby areas, and many return when the hazard recedes. Slow-onset processes, such as droughts, progressive agricultural failure and declining water availability, affect mobility differently. Rather than triggering sudden movement, they tend to set in motion a chain of economic, social and livelihood pressures that may, over time, lead people to move, or prevent them from doing so. This makes attribution considerably harder: the link between a climate stressor and a specific mobility decision is less direct, more diffuse and more easily obscured by other factors (Adger et al., 2018; JRC, 2025b). The 2023–24 El Niño illustrates this directly: flooding in East Africa generated large, rapidly documented displacement figures, while drought in Southern Africa drove comparable humanitarian need with far fewer recorded displacement events, not because fewer people were affected, but rather because drought-related mobility unfolds over longer timeframes and through multiple interacting factors.

Climate change often does not act as a direct cause of mobility but as a threat multiplier, amplifying existing fragilities (Black et al., 2011). Environment is one among multiple factors shaping migration decisions, and its influence is mediated by household characteristics, and the barriers or facilitators people face. A meta-analysis of over 50 empirical studies found strong context-dependence throughout: the level of development, the type of mobility measured, and data frequency all substantially shape what effects are found (Beine & Jeusette, 2021). The same event can produce very different mobility outcomes depending on household assets and resources, the availability of economic alternatives, social networks, governance capacity, and the adaptive strategies already in place, all factors that together determine both the incentive and the capacity to move.

Most climate-related mobility is short-distance and stays within national borders. People move to the nearest unaffected area, to family in a nearby town, or to the closest urban centre. Using census data from 72 countries spanning 1960–2016, Hoffmann et al. (2024) found that drought and aridity significantly increase internal migration. In particular, a one standard-deviation increase in aridity raised out-migration rates by up to 9.3%, with effects concentrated in agriculturally dependent areas of Sub-Saharan Africa, South Asia, the Middle East, and South America. Movement is predominantly towards urban areas within the same country. Cross-border movements are the exception; long-distance movements towards wealthy countries more so. Narratives that frame climate events as a driver of migration towards Europe are not well supported by the evidence and risk obscuring the actual humanitarian dynamics at stake (JRC, 2025d).

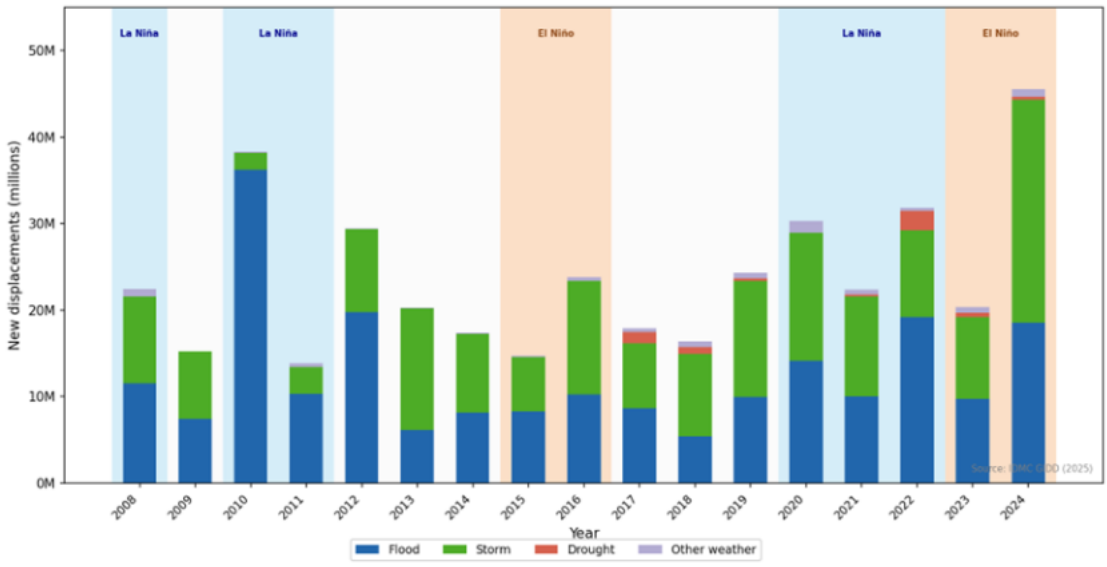
Less visible, but equally important, is the outcome where people cannot move at all. A study covering 115 countries showed that while higher temperatures raised migration rates in middle-income countries, they reduced mobility in low-income ones, where lower agricultural productivity deepens liquidity constraints and prevents households from migrating (Cattaneo and Peri 2016).

Their conclusion is direct immobility should be as concerning as mobility. In severe drought settings, precisely the conditions El Niño produces in Southern Africa and the Central American Dry Corridor, the gradual erosion of household assets can leave the poorest unable to leave even as conditions become untenable. Involuntary immobility leaves no trace in displacement data, which by definition captures only people who manage to move (JRC, 2025a; Talò & Majorano Sarapo, 2026).

6.1. Data on climate-related internal displacement

As mentioned, climate-related mobility is predominantly internal and short-distance. Figure 19 shows new climate-related internal displacements by hazard type for the period 2008–2024, as reported by the Internal Displacement Monitoring Centre (IDMC)[11], with ENSO phases indicated. Aggregate global displacement figures show no clear relationship with ENSO phase. The 2015–16 El Niño, one of the strongest on record, is associated with below-average figures: around 14.7 million new displacements in 2015. IDMC's 2016 Global Report on Internal Displacement (GRID) documents the primary reason: that event was drought-dominated, and drought-related movement is rarely recorded as displacement, it unfolds gradually, intersects with economic pressures, and may not be triggered by a discrete event. Conversely, 2024 was a record year of displacement with approximately 45.8 million new climate-related displacements, coinciding with the 2023–24 El Niño.

Figure 19. New climate-related internal displacements by hazard type, 2008–2024.



Source: JRC based on data from IDMC GIDD (2025).

Floods and storms dominate recorded new displacements in every year between 2008 and 2024. This reflects a recording asymmetry rather than a true difference in impact: sudden-onset events produce discrete, trackable displacement episodes, while slow-onset processes may generate

[11] The IDMC Global Internal Displacement Database (GIDD) is the most comprehensive global source for tracking these movements: it records new disaster-driven displacements since 2008, separates climate from conflict displacement, and provides event-level data by hazard type and country. New displacements count movements within a year, one person displaced by two separate events counts as two, capturing the frequency and intensity of hazard shocks.

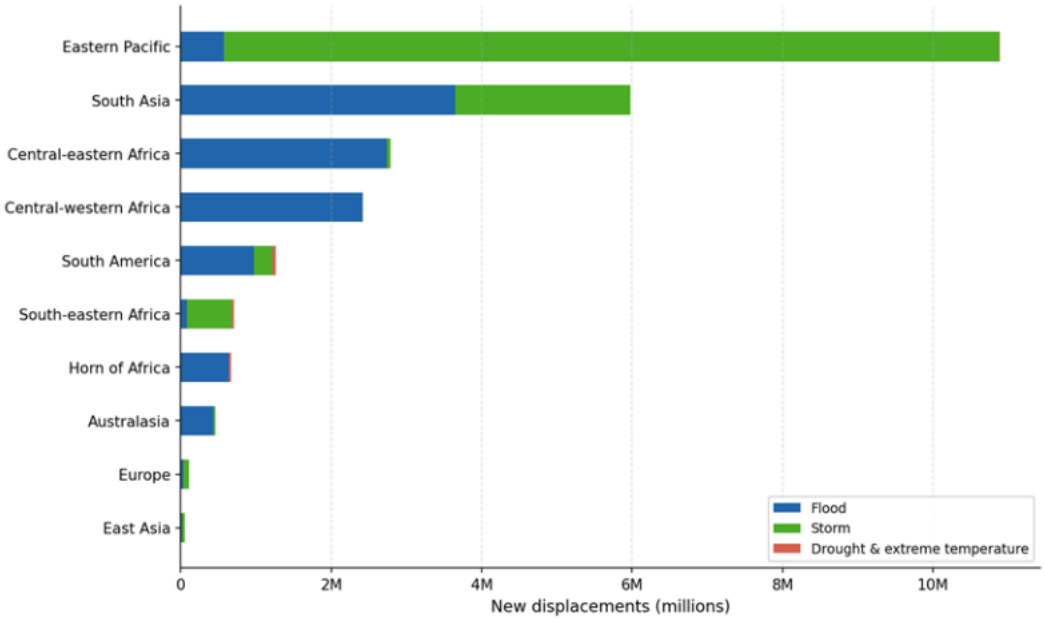
mobility that is gradual and not easily attributable to a specific climate event. Drought-driven movement tends to surface instead in internal migration or urbanisation data, or, when it crosses borders, in international migration flows, where it is difficult to disentangle from economic and other drivers. El Niño events with a large drought footprint, like 2015–16, therefore systematically appear less frequently in displacement statistics. In drought-dominated years, low displacement figures should not be read as evidence of limited impact, they are a product of the recording asymmetry described above (IDMC GRID 2016; JRC, 2025b).

6.2. Regional displacement dynamics

Displacement dynamics differ considerably between the regions of interest identified in the Anticipatory Analysis of the El Niño–Southern Oscillation report (Santini et al., 2023). In 2024, around 56,000 new displacements were recorded in East Asia, associated with events supposedly related to El Niño such as floods, storms, droughts and extreme temperatures. This contrasts sharply with the more than 10.9 million new displacements associated with such events in the Eastern Pacific. Particularly large numbers of new displacements were recorded in South Asia (6.0 million), Central-eastern Africa (2.8 million), Central-western Africa (2.4 million) and South America (1.3 million).

Figure 20 shows new displacements by region and hazard type for 2024. Flood-related displacement occurred across all major regions; South Asia, Central-eastern Africa and Central-western Africa combined accounted for more than three-quarters of flood displacements. Storm-related displacement was concentrated in the Eastern Pacific and South Asia, with the Eastern Pacific alone accounting for around three-quarters of storm displacements. Drought and extreme temperature displacements were considerably lower in absolute terms but most concentrated in South America, and South-eastern Africa, consistent with known El Niño drought signals in those regions.

Figure 20. New climate related internal displacements by region and hazard type, 2024.



Source: JRC based on data from IDMC GIDD (2025).

These regional patterns reflect the hazard structure of El Niño: flood dominance in Central Africa and South Asia, storm dominance in the Eastern Pacific and East Asia, and drought signals concentrated in South America, South-eastern Africa and the East Africa. Drought and extreme temperature displacements are considerably lower in recorded figures, as explained above. Box 1 provides a more detailed description of the impacts of the 2023-24 El Niño. In Central Africa, El Niño-associated flooding compounds pre-existing conditions of armed conflict and political instability, both amplifying exposure and undermining response and recovery capacity. This makes the region one where early preparedness investment is particularly warranted.

Box 1. The 2023-24 El Niño and different mobility outcome.

Southern Africa: drought and slow-onset mobility El Niño suppresses the November-April rainy season in Southern Africa, causing drought and possible crop failure. The 2023-24 drought was attributed primarily to El Niño, compounded by background climate warming (World Weather Attribution, 2024). The mobility response was gradual and diffuse: driven by livelihood erosion and asset depletion rather than a discrete hazard event. The International Organisation for Migration (IOM) recorded at least 202,000 displaced individuals in Zambia (84% citing drought as the primary driver, 56% female) and around 10,000 in Mozambique, both minimum estimates (IOM DTM, 2025). Involuntary immobility is likely widespread but unquantified.

East Africa: flooding and sudden-onset displacement El Niño typically intensifies the October-December short rains in the Horn of Africa (Santini et al., 2023). In late 2023, severe flooding across Somalia, Kenya and Ethiopia left over 2 million people affected and close to 1.5 million displaced by December (OCHA, 2023). Displacement was rapid, forced and short-distance, the typical pattern of sudden-onset events (JRC, 2025b). Many of those affected had endured four consecutive La Niña drought seasons between 2020 and 2023, compounding their vulnerability.

6.3. Mobility to and within Europe

El Niño's influence on European climate is indirect and less certain than in the tropics. Santini et al. (2023) identify a tendency toward warmer and drier conditions in the Mediterranean basin and wetter signals in parts of central Europe during El Niño winters, compounding existing wildfire and flood risk rather than acting as a direct hazard trigger.

Within Europe, displacement events are driven primarily by wildfires and floods. Duration of displacement is the more policy-relevant variable: how long people remain displaced depends largely on how quickly housing, services and livelihoods are restored (Talò, 2026). The EU Preparedness Union Strategy provides a directly relevant framework: anticipatory action and pre-positioned response capacity can reduce both the scale and duration of displacement following such events.

Whether El Niño-related climate stress in origin regions translates into migration flows towards Europe requires analytical care. Climate-related mobility is overwhelmingly short-distance and internal: sudden-onset events generate forced displacement that is local and temporary, while slow-onset processes may lead to local mobility, urbanisation, or immobility among the most vulnerable populations who lack the resources to move (Cattaneo & Peri, 2016; JRC, 2025b). International migration is one possible outcome but is neither automatic nor predominant. Nonetheless, 'alarmist' and 'crisis' narratives frequently portray climate change as an inevitable trigger of mass south-to-

north migration flows; such portrayals, including references to 'climate refugees' or 'climate exodus', are in key respects misaligned with the empirical evidence (JRC, 2025d). The claim that El Niño events will drive large-scale migration towards Europe is not strongly supported by existing evidence (Hoffmann et al., 2024; JRC, 2025d).

6.4. Mobility in the context of the 2026 El Niño

The 2026 El Niño, which could be of exceptional strength, heightens risk across several high-vulnerability regions simultaneously. Southern Africa faces a suppressed rainy season through the 2026-27 growing season, with populations still weakened from the 2023-24 event. Most of East Africa faces enhanced short rains and millions of existing IDPs (Internally Displaced People). Central Africa, where flood risk intersects with conflict-driven displacement, faces compounded vulnerability. The Central American Dry Corridor and parts of South and South-East Asia face drought and temperature stress. These are hazard and vulnerability signals, not mobility predictions. Compound effects are of particular concern: the 2026 event will be layered on existing displacement, conflict, and food insecurity, raising the risk of re-displacement among already vulnerable populations.

The 6-9 month El Niño forecast window does not make mobility predictable, but it does make preparedness possible. Pre-agreed funding triggers, pre-positioned supplies, cash transfer readiness and early monitoring deployment can all be activated before a crisis peaks, reducing both the scale and duration of displacement. Importantly, monitoring systems should be deployed at the point of ENSO forecast confirmation, not after displacement is already underway, given the documented challenges in capturing climate-related mobility in real time (Talò & Majorano Sarapo, 2026; JRC, 2025e).

7. Multi-lateral NGO responses, reported news, planned measures

Multi-lateral organisations, such as the United Nations (UN), are playing a crucial role in coordinating the global response to the El Niño event[12]. The UN Office for Disaster Risk Reduction (UNDRR) has issued guidelines for disaster risk reduction and management including also a use case in the context of El Niño (UNDRR, 2025). Non-governmental organisations (NGOs), such as the International Federation of Red Cross and Red Crescent Societies (IFRC), are actively engaged in disaster preparedness and response efforts[13].

Panama Channelling Controls: The Institute of Meteorology and Hydrology of Panama (IMHPA)[14] has activated severe alerts regarding extreme water depletion. The government has enacted emergency institutional coordination to ration water resources, heavily affecting draft restrictions and transit capacities through the Panama Canal.

Latin American Scaling: The IFRC[15] has mobilised its Early Action Protocols across Latin America and the Caribbean to release pre-allocated humanitarian financing ahead of the projected 2°C to 3°C Pacific temperature spikes.

Central America and Caribbean preparedness: The Caribbean Institute for Meteorology and Hydrology (CIMH) has warned the region for the threat of extreme heat during the 2026 Atlantic hurricane season, beginning in June, and for related higher energy costs. El Niño-induced drought conditions and excessive heat are expected to intensify by the end of 2026[16]. For this, CIMH urged regional governments and communities to start planning impacts. Preparatory actions have been activated in Belize for drought by the Ministry of Agriculture, Food Security and New Growth Industries, in collaboration with the National Meteorological Services and the United Nations World Food Programme (WFP). Despite not directly related to the forecasted El Niño but to a forecast indicating an increase likelihood of below-average rainfall for the upcoming season, these anticipatory actions try to respond to changes driven by ENSO patterns[17].

Dry impacts and preparedness in South America: the Ministry of Environment and Sustainable Development of Colombia, together with the Autoridad Nacional de Licencias Ambientales (ANLA) has activated a set of actions and recommendations to prepare for the possible El Niño risks of drought, water scarcity, wildfires and their impacts on water supply, food security and ecosystems[18].

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12 https://unu.edu/ehs/article/what-el-nino-and-why-does-it-matter-warming-world (13 May 2026; last access 29 May 2026)

13 https://www.ifrc.org/article/2026-el-nino-how-are-we-preparing-its-impact-latin-america-and-caribbean (28 April 2026; last access 29 May 2026)

14 https://opsaa.iica.int/en/resource-2697-panama-activates-alerts-and-preventive-measures-against-the-el-nino-phenomenon-and-climate-risks-in-2026 (12 May 2026; last access 29 May 2026)

15 https://reliefweb.int/report/world/2026-super-el-nino-how-are-we-preparing-its-impact-latin-america-and-caribbean (28 April 2026; last access 29 May 2026)

16 Caribbean urged to prepare for heat, drought and rising energy bills - Jamaica Observer (29 May 2026; last access 29 May 2026)

17 https://www.wfp.org/news/belize-triggers-anticipatory-action-support-farmers-ahead-drought-risk (8 May 2026; last access: 29 May 2026)

18 https://www.infobae.com/colombia/2026/05/29/gobierno-nacional-emitio-17-medidas-por-riesgo-de-incendios-y-escasez-de-agua-exigio-planes-hidricos-ante-emergencias-ambientales/ (29 May 2026; last access: 29 May 2026)

Impacts in southern Africa: Effect of a possible strong El Niño are feared in Zimbabwe, where the Meteorological Services Department reports an 88% to 94% chance of an El Niño event during the 2026/27 rainy season, with below-normal rainfall patterns, with risk of poor harvests and water shortages[19].

Impacts in Asia: Many Asian newspapers (e.g. [20, 21, 22]) report the WMO statements about the El Niño, which can impact with drought conditions in several ASEAN (Association of Southeast Asian Nations) countries, severely threating those which highly depend on hydropower, notably Mekong countries, Nepal, and parts of Malaysia. Agriculture could also see a strong impact, exacerbating a situation already critical due to the Persian Gulf conflict and the consequent lack of fertilizers, which cannot cross the Strait of Hormuz from the Middle East countries where they are produced. At the beginning of June, the National Centre for Hydro-Meteorological Forecasting of Vietnam forecasted above-average temperatures in the whole country, with more frequent and intense heatwaves, accompanied by a reduction of total rainfall, expected 25-50% below the long-term average. Meanwhile, it warned the risk of short and localised downpours with exceptionally high rainfall intensity, increasing the likelihood of flash floods, landslides and urban flooding[23].

Impacts in India: The Indian Meteorological Department (IMD) has started forecasting the possible effects of a strong El Niño, highlighting India's northern, western, and central regions with the highest threat of drought and, on the opposite, southeastern states with risk of heavy rainfall and devastating floods[24]. At the end of May 2026, it updated the long-range forecast for the 2026 southwest monsoon, indicating a higher probability of below-normal rainfall across the country[25]. In view of possible El Niño conditions and to address possible water stress, the government of the Indian state of Karnataka organised at the begin of May a high-level technical meeting to strengthen drought preparedness, with high priority given to groundwater recharge initiatives and long-term water conservation strategies[26].

Impacts in the Philippines: the Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA) depicted already in March the possible effects of El Niño, manifesting with heavy rainfall and floods during the habagat (western monsoon) season, especially from July to September, and then in below-average precipitation, warmer temperatures and drought in autumn,

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19 https://allafrica.com/stories/202605080389.html (8 May 2026; last access: 29 May 2026)

20 https://www.nst.com.my/world/world/2026/05/1433140/warning-super-el-nino (6 May 2026; last access: 29 May 2026)

21 https://cebudailynews.inquirer.net/725169/super-el-nino-mideast-crisis-things-arent-looking-good-for-asia (7 May 2026; last access: 29 May 2026)

22 https://www.malaymail.com/news/world/2026/05/06/super-el-nino-fears-asia-faces-energy-strain-crop-damage-and-worsening-food-insecurity/218820 (6 May 2026; last access: 29 May 2026)

23 https://english.vov.vn/en/society/el-nino-forecast-raises-risk-of-extreme-heat-and-severe-weather-in-vietnam-post1296723.vov (1 June 2026; last access 2 June 2026)

24 https://www.latestly.com/india/news/el-nino-2026-these-states-and-cities-in-india-face-highest-risk-as-historic-el-nino-prepares-to-hit-7431623.html (8 May 2026; last access: 29 May 2026)

25 https://menafn.com/1111182329/Below-Normal-Southwest-Monsoon-Likely-IMD-Forecasts-Deficient-Rainfall-Across-India (29 May 2026; last access: 2 June 2026)

26 https://theprint.in/india/karnataka-govt-convenes-meeting-on-water-security-drought-preparedness/2923837/ (6 May 2026; last access: 29 May 2026)

winter and early 2027. In the island of Cebu PAGASA would continue to coordinate with the local government units through the Office of Civil Defense (OCD) for disaster preparedness measures[27,28].

Impacts in Oceania: risk of drought, extremely hot days, wildfires are predicted by Earth Sciences New Zealand and WMO for New Zealand. This situation would likely affect groundwater recharge over winter, impacting also the growing season[29].

El Niño effects in the eastern Atlantic Ocean: Concerns about possible effects of the forecasted El Niño event are present also for Canary Islands[30].

United Kingdom Mandates: The UK's Committee on Climate Change (CCC)[31] issued its Well Adapted UK report, legally advising the government to rapidly upgrade infrastructure against impending severe heatwaves, wildfires, and flood risks aggravated by a potential Super El Niño.

Fossil Fuel Transitions: Backed by the WMO's record heat anomaly data, UN Secretary-General António Guterres issued a global directive forcing member states to fast-track the transition away from fossil fuels to prevent El Niño from acting as a "threat multiplier"[32].

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27 https://newsinfo.inquirer.net/2189947/el-nino-may-develop-in-july-but-pagasa-warns-of-intense-habagat (3 March 2026; last access: 29 May 2026)

28 https://cebudailynews.inquirer.net/730412/cebu-may-face-floods-extreme-heat-as-habagat-el-nino-overlap (22 May 2026; last access: 29 May 2026)

29 https://www.rnz.co.nz/news/weather/597102/well-and-truly-on-its-way-el-nino-very-likely-to-peak-over-new-zealand-summer (3 June 2026; last access: 3 June 2026)

30 https://diariodeavisos.elespanol.com/2026/05/el-nino-2026-efectos-canarias-alerta-nasa/ (29 May 2026; last access: 29 May 2026)

31 https://www.chathamhouse.org/2026/05/2026-super-el-nino-could-expose-gaps-uk-preparedness (22 May 2026; last access 29 May 2026)

32 https://www.bbc.com/news/articles/c203rdxkezwo (23 March 2026; last access 29 May 2026)

8. Conclusions

The Joint Research Centre will continue analysing the evolution of El Niño and the associated risks in the coming months. Hazards, such as drought, floods and wildfire, will be monitored via the Copernicus Emergency Management Service as well as the Global Disaster Awareness and Coordination System. The Anomaly Hotspots of Agricultural Production ASAP system will contribute to monitor food security, while INFORM Warning will keep assessing the humanitarian risks.

Given the inherent uncertainties and the potential unprecedented intensity this 2026/27 El Niño event may reach, it is essential to keep devoting efforts and resources to enhance understanding of the complex, interlinked risks' landscape, support preparedness, early warnings and actions.

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List of abbreviations and definitions

Abbreviation	Definition
ANLA	Autoridad Nacional de Licencias Ambientales (Colombia)
ARDL	Autoregressive Distributed Lag
ARIMA	Autoregressive Integrated Moving Average
ASEAN	Association of Southeast Asian Nations
BIC	Bayesian Information Criterium
BOM	Bureau of Meteorology (Australia)
C3S	Copernicus Climate Change Service
CCC	Committee on Climate Change (United Kingdom)
CEMS	Copernicus Emergency Management Service
CESM2	Community Earth System Model version 2
CIMH	Caribbean Institute for Meteorology and Hydrology
DWD	Deutscher Wetterdienst
ECMWF	European Centre for Medium-Range Weather Forecasts
ENSO	El Niño-Southern Oscillation
EU	European Union
GDO	Global Drought Observatory
GDP	Gross Domestic Product
GIDD	Global Internal Displacement Database

GloFAS	Global Flood Awareness System
GRID	Global Report on Internal Displacement
HRW	Hard Red Winter (type of wheat)
IDMC	Internal Displacement Monitoring Centre
IDP	Internally Displaced People
IFRC	International Federation of Red Cross and Red Crescent Societies
IMD	Indian Meteorological Department
IMHPA	Institute of Meteorology and Hydrology of Panama
INFORM	Index for Risk Management
IOM	International Organisation for Migration
JRC	Joint Research Centre (European Commission)
MAPE	Mean Absolute Percentage Error
ML	Machine Learning
NGO	Non-Governmental Organisation
NOAA	National Oceanic and Atmospheric Administration
OCD	Office of Civil Defense (Philippines)
OCHA	Office for the Coordination of Humanitarian Affairs
ONI	Oceanic Niño Index
PAGASA	Philippine Atmospheric, Geophysical and Astronomical Services Administration

SARIMAX	Seasonal Autoregressive Integrated Moving Average with Exogenous Variables
SDG	Sustainable Development Goal
SPEI	Standardised Precipitation Evapotranspiration Index
SPI	Standardized Precipitation Index
SST	Sea-Surface Temperature
UK	United Kingdom of Great Britain and Northern Ireland
UN	United Nations
UNDRR	United Nations Office for Disaster Risk Reduction
UNU-INWEH	United Nations University Institute for Water, Environment and Health
US	United States of America
WFP	World Food Programme
WMO	World Meteorological Organization

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